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CHAPTER 10

PHOTOELECTRIC EFFECT

SUGGESTED TIME: 2 PERIODS

IN THIS CHAPTER

This chapter introduces the photoelectric effect, the photon model of light and Planck's constant, leading to Einstein's photoelectric equation. It ends with the wave-particle duality of light, a cornerstone of modern physics examined at the GCE Advanced Level.

- By the end of this chapter I can describe the photoelectric effect and state the conditions under which it occurs.
- By the end of this chapter I can explain why the classical wave theory of light fails to account for the experimental observations.
- By the end of this chapter I can define a photon and calculate its energy using $E = hf$ and $E = hc/\lambda$.
- By the end of this chapter I can use Planck's constant $h = 6.63 \times 10^{-34}$ J·s correctly in calculations, including conversions between joules and electron-volts.
- By the end of this chapter I can state and apply Einstein's photoelectric equation $hf = W_0 + E_{k(max)}$.
- By the end of this chapter I can determine the work function, threshold frequency and threshold wavelength of a metal from data or from a graph.

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COURSE complete, illustrated, step by step

METHODS and worked examples

EXERCISES exam style + solutions

SUMMARY SHEET the essentials to remember

CONTENTS

1. The Photoelectric Effect: Observation and Failure of Wave Theory
2. Photons and Planck's Constant
3. Einstein's Photoelectric Equation
4. Wave-Particle Duality and Applications
5. Integration activity
6. Methods and worked examples
7. Exercises
8. Solutions to the exercises
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CHAPTER OBJECTIVES

- By the end of this chapter I can describe the photoelectric effect and state the conditions under which it occurs.
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- By the end of this chapter I can state and apply Einstein's photoelectric equation $hf = W_0 + E_{k(max)}$.
- By the end of this chapter I can determine the work function, threshold frequency and threshold wavelength of a metal from data or from a graph.
- By the end of this chapter I can interpret the graph of maximum kinetic energy of photoelectrons against frequency of incident radiation.
- By the end of this chapter I can explain the meaning of wave-particle duality and give evidence for both aspects of light.
- By the end of this chapter I can relate the photoelectric effect to applications such as photocells, solar panels and light sensors.

PREREQUISITES

- Wave properties of light: frequency, wavelength and the relation c = fλ (Lower Sixth)
- The electromagnetic spectrum and orders of magnitude of wavelengths
- Energy, work and the electron-volt as a unit of energy
- Structure of the atom: electrons, ionisation and energy levels (Lower Sixth)
- Basic graph skills: gradient, intercepts and equation of a straight line

1 THE PHOTOELECTRIC EFFECT: OBSERVATION AND FAILURE OF WAVE THEORY

In Douala, solar panels on rooftops convert sunlight directly into electricity, and automatic street lights in Yaoundé switch on by themselves at dusk. Yet for decades, physicists could not explain why shining a faint blue light on a metal ejects electrons instantly, while an intense red light — however bright — ejects nothing at all. This puzzle, which classical wave theory completely failed to solve, forced Einstein to propose that light itself arrives in tiny packets of energy called photons. In this chapter we uncover how this revolutionary idea explains the photoelectric effect and opened the door to quantum physics.

The problem of this section. We begin with the experiment itself: what exactly happens when light falls on a metal? We describe the apparatus, state the four experimental laws that any correct theory must explain, and then show, point by point, why the classical wave picture of light — so successful for interference and diffraction — collapses completely here. Finally we introduce the **stopping potential**, the experimental tool that lets us measure the energy of the ejected electrons.

1.1 Observation: Light Can Tear Electrons Out of a Metal

Intuition first. Inside a metal, some electrons (the conduction electrons) move almost freely between the atoms, but they are still bound to the metal as a whole: to escape through the surface, an electron must receive a minimum amount of energy. Heating the metal can supply this energy (thermionic emission, used in old radio valves). The surprising discovery, made by Hertz in 1887 while experimenting with electromagnetic waves, and studied in detail by Hallwachs and Lenard, is that light alone, at room temperature, can also do the job — but only under very strange conditions.

DEFINITION **Photoelectric effect and photoelectrons**

The **photoelectric effect** is the emission of electrons from the surface of a metal (or, more generally, a material) when electromagnetic radiation of sufficiently high frequency falls on it.
The emitted electrons are called **photoelectrons**. They are perfectly ordinary electrons; the prefix “photo” simply recalls that they were set free by light.

WARNING **Photoelectrons are not a new particle**

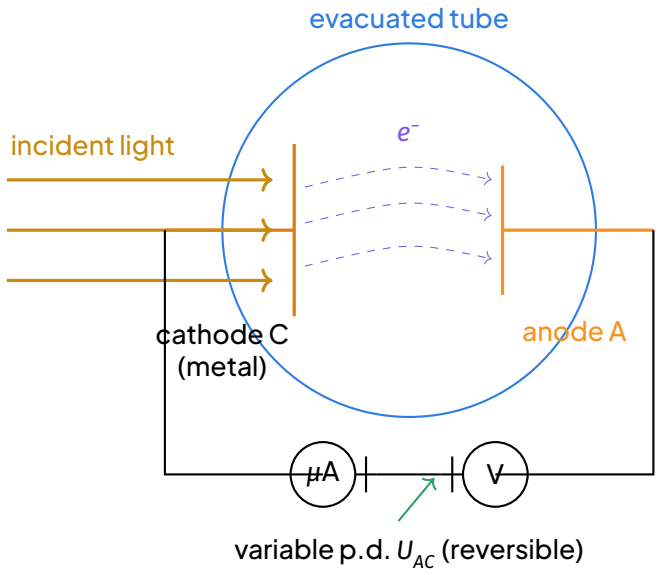
A photoelectron has exactly the same charge $-e = -1.60 \times 10^{-19}$ C and the same mass $m_e = 9.11 \times 10^{-31}$ kg as any other electron. Only its origin (ejection by light) is special.

1.2 The Experimental Set-Up: The Photoelectric Cell

To study the effect quantitatively we use a **photoelectric cell** (photoemissive cell). Its essential

parts are:

- an **evacuated glass tube**, so that the ejected electrons travel without colliding with air molecules;
- a metal plate, the **cathode** (or photocathode), coated with the metal under study (e.g. zinc, sodium, caesium), on which the light falls;
- a collecting electrode, the **anode**, facing the cathode;
- an external circuit containing a **variable potential difference** (potentiometer arrangement, allowing the anode to be made positive or negative relative to the cathode), a **microammeter** (or sensitive galvanometer) to detect the tiny photoelectric current, and a voltmeter.



The photoelectric cell. Light falling on the cathode releases photoelectrons; those reaching the anode constitute the photoelectric current measured by the microammeter.

How the measurement works. When the anode is made positive with respect to the cathode ($U_{AC} > 0$), the ejected electrons are attracted to the anode and a current I flows, detected by the microammeter. The photoelectric current is simply the rate of flow of photoelectrons:

$$I = \frac{Q}{t} = \frac{Ne}{t},$$

where N is the number of photoelectrons collected in time t. Hence the current directly counts the emitted electrons: $N/t = I/e$. When U_{AC} is made large enough, every emitted electron is collected: the current then reaches a maximum value called the **saturation current** I_s, which measures the total rate of emission from the cathode.

1.3 The Four Experimental Laws of Photoelectric Emission

Careful experiments (notably by Lenard, and later the precision measurements of Millikan) established four facts that any theory of the effect must explain.

PROPERTY **Experimental laws of the photoelectric effect**

1. Threshold frequency. For each metal there exists a minimum frequency f_0, called the **threshold frequency**, below which no photoelectron is emitted, whatever the intensity of the light. The value of f_0 is characteristic of the metal (equivalently, a threshold wavelength $\lambda_0 = c/f_0$).
2. Instantaneous emission. When $f > f_0$, emission begins essentially instantaneously (delay less than 10^{-9} s), even for extremely weak light.
3. Current proportional to intensity. For $f > f_0$, the number of photoelectrons emitted per second — and therefore the saturation current — is directly proportional to the intensity of the radiation.
4. Kinetic energy depends on frequency only. The maximum kinetic energy $E_{k(\max)}$ of the photoelectrons increases with the frequency of the radiation and is completely independent of its intensity.

EXEMPLE **Threshold frequencies of some metals**

Zinc emits electrons only under ultraviolet light ($f_0 \approx 8.1 \times 10^{14}$ Hz, $\lambda_0 \approx 370$ nm), which is why a charged zinc plate under ordinary visible light keeps its charge. Sodium and caesium have lower thresholds and respond to visible light: caesium can even emit under red light. This metal-dependence of f_0 is itself a crucial clue: the threshold is a property of the material, not of the light alone.

1.4 What Classical Wave Theory Predicts — and Why It Fails

The classical picture. In the wave theory of the 19th century, light is a continuous electromagnetic wave. Its energy is spread smoothly over the whole wavefront, and the energy delivered per second to a given area is measured by the intensity. An electron at the metal surface should therefore absorb energy gradually, like a small boat rocked by ocean swells, until it has accumulated enough to escape.

From this picture, three clear predictions follow:

1. **Any frequency should work.** Given enough time, a wave of any frequency delivers arbitrarily large energy. A feeble red light, shining long enough, should eventually eject electrons. There should be no threshold frequency.
2. **A time delay should exist for weak light.** With very dim light the energy trickles in slowly, so electrons should be emitted only after a measurable delay — seconds, minutes or even hours for the faintest sources.
3. **Kinetic energy should grow with intensity.** A more intense wave carries more energy per second, so it should shake the electrons harder and eject them with greater kinetic energy,

whatever the frequency.

Observation	Wave theory prediction	Experiment
Threshold frequency f_0	Any frequency works if intense or long enough	No emission at all for $f < f_0$, however intense the light
Time delay	Long delay for weak light	Emission in less than 10^{-9} s, even for very weak light
$E_{k(\max)}$ vs intensity	$E_{k(\max)}$ increases with intensity	$E_{k(\max)}$ totally independent of intensity
$E_{k(\max)}$ vs frequency	No special role for frequency	$E_{k(\max)}$ increases linearly with f
Effect of intensity	Stronger light ⇒ more energetic electrons	Stronger light ⇒ more electrons, each with the same maximum energy

The crisis. Every single prediction fails. The most dramatic failure is the threshold: an intense red spotlight delivers far more energy per second than a faint ultraviolet lamp, yet the red light ejects nothing while the ultraviolet acts instantly. A simple estimate makes the absurdity concrete: treating the electron as absorbing energy from an area of atomic size, classical theory predicts delays of minutes or hours for weak sources — experiment gives less than a nanosecond. Classical physics, triumphant in explaining interference and diffraction, was bankrupt before this experiment. A radically new model was needed: Einstein's photon hypothesis (developed in the next section).

WARNING **The classic trap: "brighter red light will eventually work"**

A very common mistake is to believe that increasing the intensity of red light (or waiting longer) will eventually cause emission from a metal whose threshold lies in the blue or ultraviolet. This is false. If $f < f_0$, not a single photoelectron is emitted, whatever the intensity and however long you wait. Intensity controls how many electrons are emitted per second; frequency alone decides whether emission occurs at all and how energetic the electrons are.

1.5 The Stopping Potential: Measuring the Electron Energy

Idea. How can we measure the maximum kinetic energy of the photoelectrons? By making them climb an electrostatic "hill". If we reverse the supply so that the anode becomes negative relative to the cathode ($U_{AC} < 0$), the electric field opposes the motion of the electrons: only the fastest ones reach the anode. As we make the anode more and more negative, the current falls, and at one precise voltage it becomes exactly zero.

DEFINITION **Stopping potential**

The **stopping potential** V_s (also written U_0) is the magnitude of the reverse (retarding) potential difference between anode and cathode for which the photoelectric current just

falls to zero: even the most energetic photoelectrons are turned back before reaching the anode.

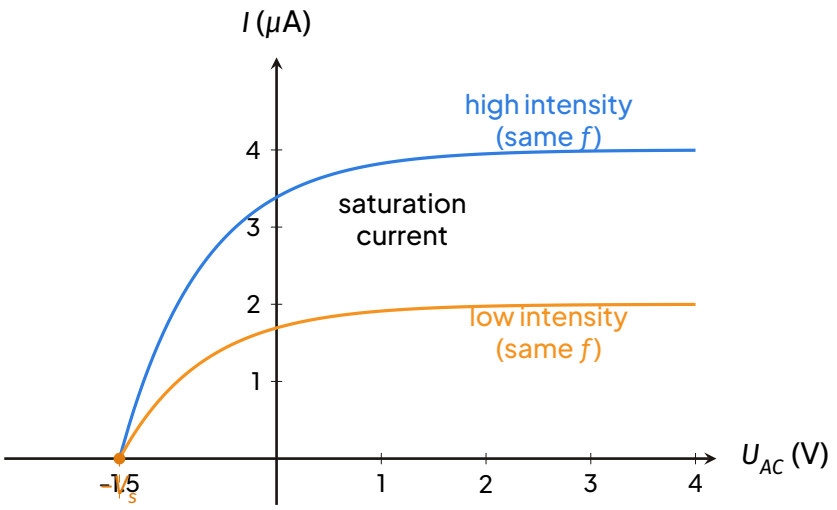
Derivation (examinable). Apply the work–energy theorem to the fastest photoelectron, emitted from the cathode with kinetic energy $E_{k(\max)}$ and travelling against a retarding p.d. of magnitude V_s. The work done by the electric force on the electron (charge $-e$) over the retarding voltage is $W = -eV_s$. At the stopping potential the electron arrives at the anode with zero speed, so

$$\Delta E_k = W \quad \Longrightarrow \quad 0 - E_{k(\max)} = -eV_s,$$

hence

$$\boxed{E_{k(\max)} = e\,V_s}$$

with $e = 1.60 \times 10^{-19}$ C. This is why energies in atomic physics are often quoted in **electronvolts**: 1 eV = 1.60×10^{-19} J is the kinetic energy gained by one electron accelerated through 1 V.



Photoelectric current versus applied voltage for two intensities of the same monochromatic light. The stopping potential V_s is the same for both curves (it depends on frequency, not intensity), while the saturation current doubles with intensity.

The graph summarises the whole section beautifully: both curves start at the same stopping potential $-V_s$ (law 4: energy depends on frequency only), but the brighter light gives a larger saturation current (law 3: more electrons per second).

METHOD **Using the stopping potential to find $E_{k(\max)}$**

1. Illuminate the cathode with monochromatic light of known frequency f (check $f > f_0$).
2. Reverse the supply and increase the retarding voltage slowly until the microammeter

reads exactly zero.

3. Read the voltmeter: its magnitude is the stopping potential V_s.
4. Compute the maximum kinetic energy: $E_{k(\max)} = eV_s$ (in joules), or directly $E_{k(\max)} = V_s$ in electronvolts.
5. Optionally, deduce the maximum speed from $E_{k(\max)} = \frac{1}{2}m_e v_{\max}^2$, i.e. $v_{\max} = \sqrt{2eV_s/m_e}$.

WORKED EXAMPLE 1 **Stopping potential of a sodium cathode**

Sodium is illuminated with ultraviolet light and emits photoelectrons. The photoelectric current is found to vanish when the retarding potential difference reaches $V_s = 1.85\ \text{V}$.
(a) Calculate the maximum kinetic energy of the photoelectrons, in joules and in electronvolts.
(b) Calculate the maximum speed of the photoelectrons.
(c) The intensity of the ultraviolet source is now doubled. What happens to the stopping potential and to the saturation current?

SOLUTION

Solution.
(a) At the stopping potential, $E_{k(\max)} = eV_s$:

$$E_{k(\max)} = 1.60 \times 10^{-19} \times 1.85 = 2.96 \times 10^{-19}\ \text{J}.$$

In electronvolts this is simply $E_{k(\max)} = 1.85\ \text{eV}$.
(b) From $E_{k(\max)} = \frac{1}{2}m_e v_{\max}^2$:

$$v_{\max} = \sqrt{\frac{2E_{k(\max)}}{m_e}} = \sqrt{\frac{2 \times 2.96 \times 10^{-19}}{9.11 \times 10^{-31}}} = \sqrt{6.50 \times 10^{11}} \approx 8.06 \times 10^{5}\ \text{m s}^{-1}.$$

(c) Doubling the intensity doubles the number of photons arriving per second, so the saturation current doubles. But the energy of each photoelectron depends only on the frequency, which is unchanged: the stopping potential stays at $V_s = 1.85\ \text{V}$.

WARNING **Sign conventions with the stopping potential**

The stopping potential is quoted as a magnitude (a positive number of volts), but it corresponds to a reverse voltage: the anode is negative relative to the cathode, $U_{AC} = -V_s$. In $E_{k(\max)} = eV_s$, use $e = +1.60 \times 10^{-19}\ \text{C}$ (the elementary charge, positive). Mixing up the signs is a frequent source of error in the examination.

EXERCISE 1 **Threshold and stopping potential**

A zinc cathode ($\lambda_0 = 370$ nm) is illuminated successively by: (i) an intense red laser of wavelength 650 nm; (ii) a very weak ultraviolet lamp of wavelength 250 nm.
1. For each source, state whether photoelectrons are emitted, justifying your answer.
2. For the case where emission occurs, the stopping potential is measured as $V_s = 1.60$ V. Deduce $E_{k(\max)}$ in joules and the maximum speed of the photoelectrons.
3. The red laser's power is multiplied by 1000. Does anything change? Comment.

SOLUTION **Exercise 1**

1. (i) Red light: $f = c/\lambda = 3.00 \times 10^8/(650 \times 10^{-9}) = 4.62 \times 10^{14}$ Hz, while $f_0 = c/\lambda_0 = 3.00 \times 10^8/(370 \times 10^{-9}) = 8.11 \times 10^{14}$ Hz. Since $f < f_0$, no emission, despite the high intensity. (ii) Ultraviolet: $f = 3.00 \times 10^8/(250 \times 10^{-9}) = 1.20 \times 10^{15}$ Hz $> f_0$: emission occurs, instantly, even though the lamp is weak.
2. $E_{k(\max)} = eV_s = 1.60 \times 10^{-19} \times 1.60 = 2.56 \times 10^{-19}$ J; $v_{\max} = \sqrt{2E_{k(\max)}/m_e} = \sqrt{5.62 \times 10^{11}} \approx 7.50 \times 10^5$ m s^{-1}.
3. Nothing changes: the frequency is still below threshold, so no electron is emitted, whatever the power. Only raising the frequency above f_0 would produce emission.

KEY POINTS **Key points of this section**

- The photoelectric effect is the emission of electrons from a metal illuminated by radiation of frequency f greater than a threshold f_0 characteristic of the metal.
- Four experimental laws: threshold frequency; instantaneous emission ($< 10^{-9}$ s); photoelectric current proportional to intensity; maximum kinetic energy depending on frequency only.
- Classical wave theory predicts the opposite on every point: this crisis demanded a new model of light.
- The stopping potential V_s is the reverse voltage that just reduces the current to zero; it measures the maximum kinetic energy through $E_{k(\max)} = eV_s$.
- Intensity controls how many electrons; frequency controls whether and how fast.

2 PHOTONS AND PLANCK'S CONSTANT

In the previous section we saw that the photoelectric effect stubbornly refuses to be explained

by the wave theory of light. Three facts stood out like accusations: there is a **threshold frequency** below which no electron is ever ejected, however intense the light; the maximum kinetic energy of the emitted electrons depends only on the frequency, not on the intensity; and the emission is instantaneous, with no measurable delay even for very weak light. If light were a continuous wave spreading its energy evenly over the metal surface, none of this should happen. A new picture of light is needed. That picture was built in two revolutionary steps: Planck's quantum hypothesis (1900) and Einstein's photon model (1905).

2.1 Planck's Quantum Hypothesis (1900)

The problem Planck was trying to solve.

Around 1900, physicists were struggling to explain the spectrum of the radiation emitted by a hot body (an idealised emitter called a black body). Classical physics predicted that a hot object should emit infinitely more energy at short wavelengths — the so-called "ultraviolet catastrophe" — in flagrant contradiction with experiment. The German physicist Max Planck found a formula that fitted the experimental curves perfectly, but to derive it he had to make a desperate and, in his own view, almost absurd assumption.

DEFINITION **Planck's quantum hypothesis**

The energy of the oscillating charges that emit and absorb electromagnetic radiation is not continuous: it can only be exchanged in **discrete packets** called **quanta** (singular: quantum). Each quantum of radiation of frequency f carries the energy

$$E = hf$$

where h is a universal constant of nature called **Planck's constant**.

In other words, energy at the microscopic scale behaves like money: you can pay 100 FCFA or 200 FCFA, but never 137.4 FCFA if the smallest coin is 100 FCFA. Nature, at the atomic scale, does not allow arbitrary amounts of energy — only whole multiples of hf.

DEFINITION **Planck's constant**

Planck's constant h is the fundamental constant of quantum physics that fixes the size of the energy quantum:

$$h = 6.63 \times 10^{-34}\ \text{J s} \quad \text{(joule-second)}.$$

Its unit follows directly from $E = hf$: $[h] = \dfrac{[E]}{[f]} = \dfrac{\text{J}}{\text{Hz}} = \text{J s}$.

WARNING **Why is the quantum world invisible in daily life?**

Planck's constant is extremely small: 6.63×10^{-34} J s. A torch beam of frequency $f \approx 5 \times$

10^{14} Hz has quanta of energy $E = hf \approx 3.3 \times 10^{-19}$ J. This is so tiny that the "graininess" of light energy is completely undetectable in everyday life, exactly as we do not feel the individual atoms when we touch a smooth table. The quantum structure of energy only reveals itself at the scale of atoms and electrons.

2.2 Einstein's Photon Model (1905)

Planck had quantised the emission and absorption of light, but he still believed light itself travelled as a wave. In 1905, Albert Einstein took the idea much further to explain the photoelectric effect.

DEFINITION **The photon**

Light (and every electromagnetic radiation) consists of a stream of particles of energy called **photons**. Each photon:

- carries a quantum of energy $E = hf$;
- travels in vacuum at the speed of light $c = 3.0 \times 10^8\ \mathrm{m\,s^{-1}}$;
- has zero rest mass and carries no electric charge;
- interacts with matter "all or nothing": it is either completely absorbed or not absorbed at all.

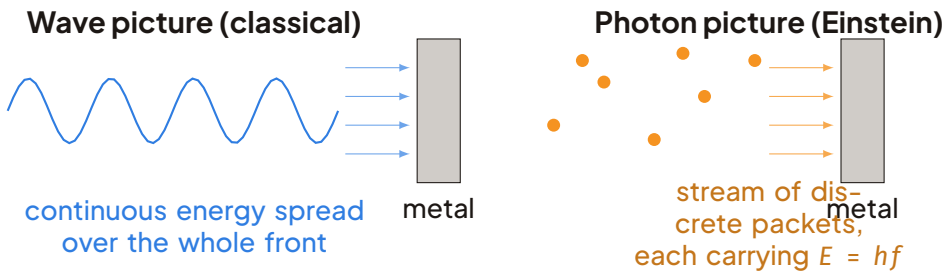
PROPERTY **Energy of a photon**

The energy of a photon of frequency f and wavelength λ (in vacuum) is

$$E = hf = \frac{hc}{\lambda}, \qquad \text{since } c = f\lambda.$$

With $h = 6.63 \times 10^{-34}$ J s and $c = 3.0 \times 10^8\ \mathrm{m\,s^{-1}}$, the product $hc = 1.99 \times 10^{-25}$ J m.

This relation shows immediately that short wavelengths mean energetic photons: ultraviolet photons are more energetic than visible photons, which are themselves more energetic than infrared photons. This is exactly what the photoelectric experiments demanded.



Two pictures of light striking a metal surface. Left: the classical wave spreads its energy continuously over the surface. Right: Einstein's photons arrive as discrete, localised packets of energy $E = hf$.

2.3 The Electron-Volt: A Convenient Unit

Photon energies are of the order of 10^{-19} J. Writing such numbers is painful and error-prone, so atomic and quantum physicists use a much more convenient unit.

DEFINITION **The electron-volt**

One **electron-volt** (symbol: eV) is the kinetic energy gained by an electron accelerated from rest through a potential difference of one volt:

$$1\,\text{eV} = e \times 1\,\text{V} = 1.60 \times 10^{-19}\,\text{J}.$$

METHOD **Converting photon energies between joules and electron-volts**

1. Compute the energy in joules: $E = \dfrac{hc}{\lambda}$ with λ in metres.
2. To convert joules → electron-volts, divide by 1.60×10^{-19}:

$$E\,(\text{eV}) = \frac{E\,(\text{J})}{1.60 \times 10^{-19}\,\text{J/eV}}.$$

3. To convert electron-volts → joules, multiply by 1.60×10^{-19}.
4. Quick check: a visible photon must give an answer of a few eV. If you find 10^{15} eV or 10^{-30} eV, a unit conversion went wrong.

WARNING **The classic trap: wavelength not in metres**

In $E = \frac{hc}{\lambda}$, the wavelength λ must be in metres. GCE questions routinely give λ in nanometres. Before any calculation, convert:

$$1\,\text{nm} = 10^{-9}\,\text{m}, \qquad 1\,\text{cm} = 10^{-2}\,\text{m}.$$

Forgetting this conversion introduces an error of 10^9 — your photon becomes absurdly energetic, and the mark is lost.

2.4 Photon Energies Across the Spectrum: Worked Examples

WORKED EXAMPLE 2 **Energy of a red photon and a violet photon**

Statement. Calculate, in joules and in electron-volts, the energy of (a) a red photon of wavelength $\lambda_R = 700$ nm; (b) a violet photon of wavelength $\lambda_V = 400$ nm. Take $h = 6.63 \times 10^{-34}$ J s, $c = 3.0 \times 10^8$ m s^{-1}, 1 eV $= 1.60 \times 10^{-19}$ J.

SOLUTION

Solution.
(a) Red photon. Convert the wavelength: $\lambda_R = 700\,\text{nm} = 7.00 \times 10^{-7}$ m.

$$E_R = \frac{hc}{\lambda_R} = \frac{(6.63 \times 10^{-34})(3.0 \times 10^8)}{7.00 \times 10^{-7}}$$
$$= \frac{1.989 \times 10^{-25}}{7.00 \times 10^{-7}} = 2.84 \times 10^{-19}\,\text{J}.$$

Converting to electron-volts:

$$E_R = \frac{2.84 \times 10^{-19}}{1.60 \times 10^{-19}} = 1.8\,\text{eV}.$$

(b) Violet photon. $\lambda_V = 400\,\text{nm} = 4.00 \times 10^{-7}$ m.

$$E_V = \frac{1.989 \times 10^{-25}}{4.00 \times 10^{-7}} = 4.97 \times 10^{-19}\,\text{J} = 3.1\,\text{eV}.$$

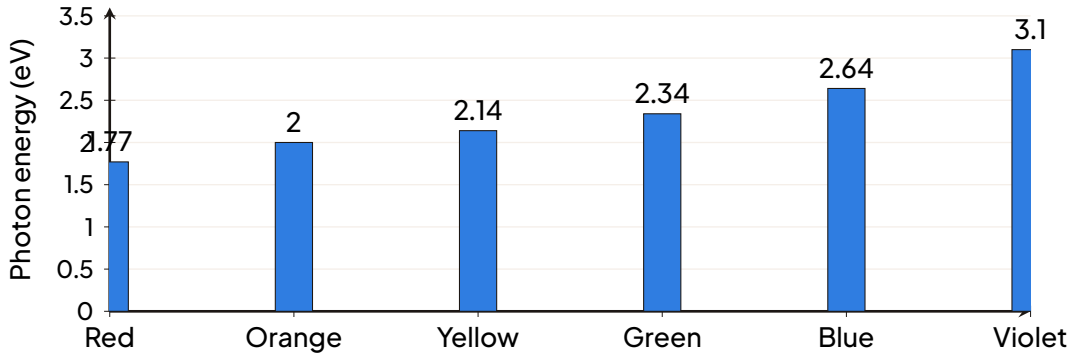
Conclusion: visible photons carry energies of a few electron-volts, increasing from red ($\approx$ 1.8 eV) to violet ($\approx$ 3.1 eV). Violet photons are nearly twice as energetic as red ones — which is why violet light can eject electrons from metals on which red light has no effect.

EXEMPLE **Orders of magnitude across the electromagnetic spectrum**

Using $E = hc/\lambda$:

- Infrared, $\lambda = 1.0 \times 10^{-6}$ m: $E \approx 2.0 \times 10^{-19}$ J ≈ 1.2 eV — usually too weak to cause photoemission from most metals.
- Visible, $\lambda \approx 400$–700 nm: $E \approx 1.8$–3.1 eV.
- Ultraviolet, $\lambda = 200$ nm: $E \approx 9.9 \times 10^{-19}$ J ≈ 6.2 eV — energetic enough for photoemission from many metals.

The energy of one photon is fixed by its frequency alone; making the light brighter does not make individual photons more energetic.



Photon energies across the visible spectrum, from red ($\lambda \approx 700$ nm, 1.77 eV) to violet ($\lambda \approx 400$ nm, 3.10 eV). Energy grows steadily as wavelength decreases.

2.5 The Photon Interpretation of Intensity

If light is a hail of photons, what does its intensity mean? In the wave picture, intensity was the energy carried per unit area per unit time by the wave. In the photon picture, the same energy is transported by the photons, so:

PROPERTY **Intensity in the photon model**

The intensity I of a monochromatic beam (frequency f) equals the number N of photons crossing a unit area per unit time, multiplied by the energy of one photon:

$$I = N \times hf \quad \Longleftrightarrow \quad N = \frac{I}{hf}.$$

For a source of power P illuminating an area A, the number of photons arriving per second on that area is

$$n = \frac{P}{hf} = \frac{P}{E_{\text{photon}}}.$$

WARNING **Exam tip: photons per second from power**

GCE questions very often give the power (or intensity and area) of a light source and ask for the number of photons emitted or absorbed per second. The key relation is

$$\text{number of photons per second} = \frac{\text{power}}{\text{energy of one photon}} = \frac{P}{hf} = \frac{P\lambda}{hc}.$$

Compute E_{photon} first (in joules!), then divide the power by it.

WORKED EXAMPLE 3 **Counting photons from a lamp**

Statement. A small sodium lamp emits yellow light of wavelength $\lambda = 589$ nm with a useful light power $P = 2.0$ W. Calculate the number of photons emitted per second. ($h = 6.63 \times 10^{-34}$ J s, $c = 3.0 \times 10^{8}$ m s^{-1}.)

SOLUTION

Solution.
Energy of one photon (with $\lambda = 589 \times 10^{-9}$ m):

$$E_{\text{photon}} = \frac{hc}{\lambda} = \frac{(6.63 \times 10^{-34})(3.0 \times 10^{8})}{589 \times 10^{-9}} = 3.38 \times 10^{-19}\ \text{J}.$$

Number of photons per second:

$$n = \frac{P}{E_{\text{photon}}} = \frac{2.0}{3.38 \times 10^{-19}} \approx 5.9 \times 10^{18}\ \text{photons per second}.$$

Even a modest lamp emits nearly six billion billion photons every second — which is why light usually looks continuous.

2.6 How the Photon Model Explains the Photoelectric Effect

We can now close the loop and see how naturally the photon model dissolves the three paradoxes of the previous section.

1. The threshold frequency. In the photon picture, an electron is ejected in a one photon – one electron collision: a single photon is absorbed completely by a single electron. The electron can escape the metal only if the photon's energy hf is at least equal to the minimum work needed to extract it, the **work function** W_0 of the metal (studied in detail in the next section):

$$hf \geq W_0 \quad \Longleftrightarrow \quad f \geq f_0 = \frac{W_0}{h}.$$

If $f < f_0$, each photon is individually too weak. Increasing the intensity merely sends more insufficient photons — like throwing millions of ping-pong balls at a door: none can break it, however many you throw. One energetic "bullet" (a photon with $f \geq f_0$) succeeds where an army of weak ones fails.

2. Instantaneous emission. The energy is delivered in a single, localised packet. As soon as a photon with $f \geq f_0$ hits a suitable electron, the electron is ejected immediately — there is no need to accumulate energy gradually from a spread-out wave front. Even at very low intensity, the first photon that arrives can eject an electron, with no measurable delay (less than about 10^{-9} s experimentally).

3. Kinetic energy depends on frequency, not intensity. Since the interaction is one photon for one electron, the surplus energy $hf - W_0$ becomes the electron's kinetic energy: it depends on f alone. Raising the intensity at fixed f increases the number of photons, hence the number of ejected electrons (the current), but not their individual energies.

WORKED EXAMPLE 4 **Threshold frequency of sodium**

Statement. The work function of sodium is $W_0 = 2.3$ eV. (a) Calculate the threshold frequency f_0 and the corresponding threshold wavelength λ_0 of sodium. (b) State, with justification, whether green light of wavelength 550 nm can eject electrons from sodium. ($h = 6.63 \times 10^{-34}$ J s, $c = 3.0 \times 10^8$ m s^{-1}, 1 eV $= 1.60 \times 10^{-19}$ J.)

SOLUTION

Solution.
(a) Convert the work function to joules:

$$W_0 = 2.3 \times 1.60 \times 10^{-19} = 3.68 \times 10^{-19}\ \text{J}.$$

Threshold frequency:

$$f_0 = \frac{W_0}{h} = \frac{3.68 \times 10^{-19}}{6.63 \times 10^{-34}} = 5.6 \times 10^{14}\ \text{Hz}.$$

Threshold wavelength:

$$\lambda_0 = \frac{c}{f_0} = \frac{3.0 \times 10^8}{5.6 \times 10^{14}} = 5.4 \times 10^{-7}\ \text{m} = 540\ \text{nm}.$$

(b) Green light has $\lambda = 550$ nm $> \lambda_0 = 540$ nm, i.e. $f < f_0$. Its photon energy is

$$E = \frac{hc}{\lambda} = \frac{1.989 \times 10^{-25}}{5.50 \times 10^{-7}} = 3.62 \times 10^{-19}\ \text{J} = 2.3\ \text{eV},$$

marginally below the work function. Therefore green light of 550 nm cannot eject electrons from sodium, however intense it may be.

KEY POINTS **Key points of this section**

- Planck (1900): energy is exchanged in quanta $E = hf$, with $h = 6.63 \times 10^{-34}$ J s.
- Einstein (1905): light itself is a stream of photons, each of energy $E = hf = \frac{hc}{\lambda}$, travelling

at speed c.

- 1 eV = 1.60×10^{-19} J; visible photons carry a few eV.
- Intensity = (photons per unit area per unit time) × (photon energy); photons per second = $P/(hf)$.
- The one photon – one electron picture explains the threshold frequency $f_0 = W_0/h$ and the instantaneous character of photoemission.

EXERCISE 2 **Photons from a UV source**

An ultraviolet source emits light of wavelength $\lambda = 250$ nm with a power of 0.50 W.

1. Calculate the energy of one photon, in joules and in electron-volts.
2. How many photons does the source emit per second?
3. The work function of zinc is $W_0 = 4.3$ eV. Can this radiation eject electrons from a zinc plate? Justify.

SOLUTION **Exercise 1**

1. $\lambda = 250 \times 10^{-9}$ m.

$$E = \frac{hc}{\lambda} = \frac{1.989 \times 10^{-25}}{2.50 \times 10^{-7}} = 7.96 \times 10^{-19}\ \text{J} = \frac{7.96 \times 10^{-19}}{1.60 \times 10^{-19}} = 5.0\ \text{eV}.$$

2. $n = \frac{P}{E} = \frac{0.50}{7.96 \times 10^{-19}} \approx 6.3 \times 10^{17}$ photons per second. 3. Yes: $E = 5.0$ eV > $W_0 = 4.3$ eV, so each photon has enough energy to extract an electron from zinc (the surplus 0.7 eV becomes kinetic energy).

DID YOU KNOW? **A Nobel Prize for the photon**

It was for his explanation of the photoelectric effect — and not for relativity — that Albert Einstein received the Nobel Prize in Physics in 1921. The idea that light comes in particles was so shocking that even Planck initially opposed it. Today, photons are the workhorses of modern technology: solar panels powering homes across Cameroon, photodiodes in fibre-optic communications, and the sensors of every smartphone camera all rely on photon–electron interactions.

3 EINSTEIN'S PHOTOELECTRIC EQUATION

In the previous section we established that light behaves as a stream of energy quanta called **photons**, each carrying an energy $E = hf = \frac{hc}{\lambda}$, where $h = 6.63e-34$ J s is Planck's constant. We now use this revolutionary idea to do what the wave theory could never do: explain the photoelectric effect quantitatively. Einstein's photoelectric equation, published in 1905, is nothing more than the conservation of energy applied to the interaction between one photon and one electron in a metal. It is one of the most frequently examined relations at the GCE Advanced Level, so we shall build it carefully, term by term.

3.1 The Work Function and the Threshold Frequency

Why an electron needs energy to escape.

Intuition first. Imagine a child at the bottom of a deep well in the village. To climb out, the child must spend energy working against gravity; the deeper the well, the more energy is needed. An electron inside a metal is in a similar situation: it is attracted by the positive ions of the metallic lattice, so it sits in a kind of "potential well". To tear it away from the metal surface, energy must be supplied to it. Different metals hold their electrons with different strengths, just as wells have different depths.

DEFINITION **Work function**

The **work function** W_0 of a metal is the minimum energy required to extract an electron from the surface of that metal. It is a property of the metal alone (not of the light used). It is related to the **threshold frequency** f_0 and the **threshold wavelength** λ_0 by

$$W_0 = hf_0 = \frac{hc}{\lambda_0}.$$

DEFINITION **Threshold frequency and threshold wavelength**

The **threshold frequency** f_0 is the lowest frequency of radiation that can eject electrons from a given metal. The **threshold wavelength** $\lambda_0 = c/f_0$ is the longest wavelength that can do so. Radiation with $f < f_0$ (equivalently $\lambda > \lambda_0$) can never cause photoemission, however intense it may be.

EXEMPLE **Typical values for common metals**

The table below gives approximate values frequently used in GCE problems. Work functions are often quoted in electronvolts: $1\,\text{eV} = 1.60 \times 10^{-19}$ J.

Metal	W_0 (eV)	f_0 (Hz)	λ_0 (nm)
Sodium (Na)	2.3	5.6×10^{14}	540
Calcium (Ca)	2.9	7.0×10^{14}	430
Zinc (Zn)	4.3	1.0×10^{15}	290

Notice that sodium responds to visible light (green), whereas zinc needs ultraviolet radiation: this is why zinc plates are used in classic demonstrations with UV lamps.

WARNING **The work function is a MINIMUM**

W_0 is the energy needed to extract the least tightly bound electrons, those right at the surface. Electrons lying deeper inside the metal need more energy to escape. This is why, as we shall see, the kinetic energy computed from Einstein's equation is a maximum.

3.2 Statement and Derivation of Einstein's Equation

One photon, one electron.

Einstein's key assumption is that the interaction is **individual**: a single photon is completely absorbed by a single electron. The photon ceases to exist and hands over all of its energy hf to the electron in one indivisible transaction. There is no gradual accumulation of energy, which immediately explains why emission is instantaneous and why weak light below the threshold produces nothing.

The electron then spends part of this energy to escape from the metal (at least W_0), and whatever remains appears as kinetic energy.

PROPERTY **Einstein's photoelectric equation**

For monochromatic radiation of frequency f incident on a metal of work function W_0:

$$\boxed{hf = W_0 + E_{k(\max)}}$$

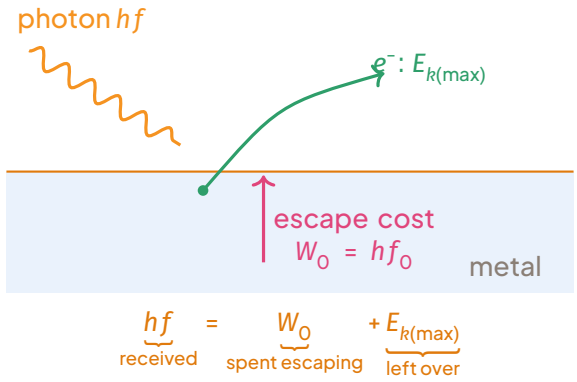
where $E_{k(\max)} = \frac{1}{2} m_e v_{\max}^2$ is the maximum kinetic energy of the emitted photoelectrons. Equivalently, using the stopping potential V_s (the reverse voltage that just stops the most energetic electrons), $E_{k(\max)} = eV_s$, so

$$hf = W_0 + eV_s.$$

This equation is a direct statement of the conservation of energy. The derivation (energy balance) is examinable and you must be able to reproduce it in words.

Derivation (energy balance).

- **Energy received:** the electron absorbs one photon of energy hf.
- **Energy spent:** to leave the metal, the electron must do work at least equal to W_0 against the attractive forces of the lattice.
- **Energy remaining:** by conservation of energy, the surplus is carried away as kinetic energy: $E_k = hf - W_0$. For a surface electron (minimum possible expenditure W_0), this kinetic energy is maximal, hence $E_{k(\max)} = hf - W_0$.



Energy balance for one absorbed photon: its energy is split between the work needed to escape the metal and the kinetic energy of the emitted electron.

3.3 Condition for Photoemission

Since kinetic energy can never be negative, emission is only possible if

$$E_{k(\max)} = hf - W_0 \geq 0 \iff hf \geq W_0 \iff \boxed{f \geq f_0 \quad \text{or equivalently} \quad \lambda \leq \lambda_0.}$$

At the exact threshold ($f = f_0$), the electron is ejected with zero kinetic energy: it just barely escapes. Below the threshold, each photon simply does not carry enough energy, and because absorption is all-or-nothing, no electron is ever emitted — even if billions of photons arrive every second.

WARNING **Intensity versus frequency — do not mix them up!**

This is the classic GCE trap:

- Increasing the intensity of the light (with $f > f_0$) means more photons per second, hence more electrons emitted per second (a larger photoelectric current). The kinetic energy of each electron is unchanged.
- Increasing the frequency means more energetic photons, hence more energetic electrons (larger $E_{k(\max)}$ and larger stopping potential). The number of electrons is es-

sentially unchanged.

Intensity controls the number; frequency controls the energy. If $f < f_0$, no intensity, however large, can produce emission.

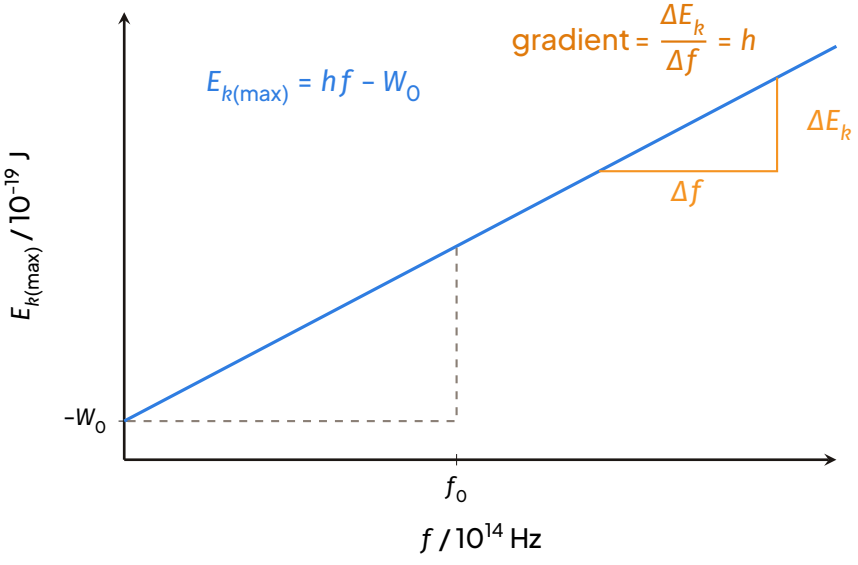
WARNING $E_{k(\max)}$ **is a MAXIMUM**

Einstein's equation gives the kinetic energy of the most energetic electrons only — those ejected from the very surface. Electrons from deeper layers lose extra energy in collisions on their way out and emerge with less kinetic energy, anywhere between 0 and $E_{k(\max)}$. Never write "all photoelectrons have kinetic energy $hf - W_0$".

3.4 The Graph of $E_{k(\max)}$ against f

Rewriting the equation as $E_{k(\max)} = hf - W_0$, we recognise the equation of a straight line $y = mx + c$ with:

- **gradient** = h (Planck's constant) — the same for all metals;
- **intercept on the frequency axis** = f_0 (the threshold frequency, where $E_{k(\max)} = 0$);
- **intercept on the energy axis** = $-W_0$.



Maximum kinetic energy versus frequency (here for sodium). The line cuts the frequency axis at f_0 and the energy axis at $-W_0$; its gradient is Planck's constant h.

METHOD **Extracting h, f_0 and W_0 from the graph**

1. Plot $E_{k(\max)}$ (in joules) on the vertical axis against f (in hertz) on the horizontal axis and draw the best straight line.
2. Gradient: pick two well-separated points on the line and compute $h = \dfrac{\Delta E_{k(\max)}}{\Delta f}$. Watch the powers of ten on the axes!
3. Threshold frequency: read f_0 where the line crosses the f-axis ($E_{k(\max)} = 0$).
4. Work function: either read the magnitude of the vertical intercept, $W_0 = |-W_0|$, or compute $W_0 = hf_0$. Convert to eV by dividing by 1.60×10^{-19} if required.
5. Threshold wavelength: $\lambda_0 = c/f_0$.

3.5 Link with the Stopping Potential

In the photocell experiment, we apply a reverse (retarding) voltage between the emitter and the collector and increase it until even the fastest electrons are turned back: the current falls to zero. The work done by the retarding field on an electron of charge e over the **stopping potential** V_s equals the maximum kinetic energy:

$$eV_s = E_{k(\max)} = hf - W_0 \qquad \Longrightarrow \qquad V_s = \frac{h}{e}f - \frac{W_0}{e}.$$

A graph of V_s against f is therefore also a straight line, of gradient h/e, with the same frequency intercept f_0. This was the basis of Millikan's famous experiments which measured h and confirmed Einstein's theory.

WORKED EXAMPLE 5 **Photoelectrons from a sodium surface**

Statement. Light of wavelength $\lambda = 400\,\text{nm}$ falls on a sodium surface of work function $W_0 = 2.28\,\text{eV}$. Take $h = 6.63e-34\,\text{J s}$, $c = 3.00e8\,\text{m/s}$, $e = 1.60e-19\,\text{C}$. Determine: (a) the threshold frequency and threshold wavelength of sodium; (b) the maximum kinetic energy of the photoelectrons, in joules and in eV; (c) the stopping potential.

SOLUTION

Solution. (a) Convert the work function to joules:

$$W_0 = 2.28 \times 1.60 \times 10^{-19} = 3.65 \times 10^{-19}\,\text{J}.$$

Threshold frequency:

$$f_0 = \frac{W_0}{h} = \frac{3.65 \times 10^{-19}}{6.63 \times 10^{-34}} = 5.50 \times 10^{14}\,\text{Hz}.$$

Threshold wavelength:

$$\lambda_0 = \frac{c}{f_0} = \frac{3.00 \times 10^8}{5.50 \times 10^{14}} = 5.45 \times 10^{-7}\ \text{m} = 545\ \text{nm}.$$

Since $\lambda = 400$ nm $< \lambda_0$, emission does occur.
(b) Photon energy:

$$hf = \frac{hc}{\lambda} = \frac{6.63 \times 10^{-34} \times 3.00 \times 10^8}{4.00 \times 10^{-7}} = 4.97 \times 10^{-19}\ \text{J}.$$

Einstein's equation:

$$E_{k(\max)} = hf - W_0 = 4.97 \times 10^{-19} - 3.65 \times 10^{-19} = 1.32 \times 10^{-19}\ \text{J} = 0.83\ \text{eV}.$$

(c) Stopping potential:

$$V_s = \frac{E_{k(\max)}}{e} = \frac{1.32 \times 10^{-19}}{1.60 \times 10^{-19}} = 0.83\ \text{V}.$$

Note the shortcut: when $E_{k(\max)}$ is expressed in eV, the stopping potential in volts has the same numerical value.

WORKED EXAMPLE 6 **Determining Planck's constant from experimental data**

Statement. In a photoelectric experiment with a calcium surface, the stopping potentials measured for two frequencies are: $V_{s1} = 0.50$ V at $f_1 = 8.0e14$ Hz and $V_{s2} = 1.60$ V at $f_2 = 1.1e15$ Hz. Deduce (a) Planck's constant, (b) the work function and the threshold frequency of calcium.

SOLUTION

Solution. (a) From $eV_s = hf - W_0$, subtracting the two measurements eliminates W_0:

$$e(V_{s2}-V_{s1}) = h(f_2-f_1) \quad \Longrightarrow \quad h = \frac{e(V_{s2}-V_{s1})}{f_2-f_1} = \frac{1.60 \times 10^{-19} \times (1.60-0.50)}{1.1 \times 10^{15} - 8.0 \times 10^{14}} = \frac{1.76 \times 10^{-19}}{3.0 \times 10^{14}} = 5.9{\times}10^{-34}\ \text{J s}.$$

(b) Using the first measurement:

$$W_0 = hf_1 - eV_{s1} = (5.9{\times}10^{-34}{\times}8.0{\times}10^{14}) - (1.60{\times}10^{-19}{\times}0.50) = 4.7{\times}10^{-19} - 0.80{\times}10^{-19} = 3.9{\times}10^{-19}\ \text{J} \approx 2.5\ \text{eV},$$

and

$$f_0 = \frac{W_0}{h} = \frac{3.9 \times 10^{-19}}{5.9 \times 10^{-34}} \approx 6.7 \times 10^{14}\ \text{Hz}.$$

(Small discrepancies with accepted values come from the rounded data — a good lesson in experimental uncertainty.)

EXERCISE 3 **GCE practice**

Ultraviolet radiation of frequency 1.2*e*15 Hz is incident on a zinc plate of work function 4.3 eV.

1. Show that photoemission occurs and calculate the threshold wavelength of zinc.
2. Calculate the maximum kinetic energy of the emitted electrons in eV.
3. Calculate the stopping potential and the maximum speed of the photoelectrons (m_e = 9.11*e* − 31 kg).
4. The intensity of the radiation is now doubled, the frequency being unchanged. State what happens to the answers of parts 2 and 3 and to the photoelectric current.

SOLUTION **Exercise 1**

1. $W_0 = 4.3 \times 1.60 \times 10^{-19} = 6.88 \times 10^{-19}$ J; $f_0 = W_0/h = 1.04 \times 10^{15}$ Hz < 1.2×10^{15} Hz, so emission occurs. $\lambda_0 = c/f_0 = 2.9 \times 10^{-7}$ m = 290 nm.
2. $hf = 6.63 \times 10^{-34} \times 1.2 \times 10^{15} = 7.96 \times 10^{-19}$ J = 4.97 eV; $E_{k(\max)} = 4.97 - 4.3 = 0.67$ eV.
3. $V_s = 0.67$ V; $E_{k(\max)} = 0.67 \times 1.60 \times 10^{-19} = 1.07 \times 10^{-19}$ J, so

$$v_{\max} = \sqrt{\frac{2E_{k(\max)}}{m_e}} = \sqrt{\frac{2 \times 1.07 \times 10^{-19}}{9.11 \times 10^{-31}}} \approx 4.9 \times 10^5 \text{ m s}^{-1}.$$

4. $E_{k(\max)}$, V_s and $v_{\max}$ are unchanged (frequency is unchanged); the photoelectric current doubles because twice as many photons eject twice as many electrons per second.

KEY POINTS **Key points**

- Work function: $W_0 = hf_0 = hc/\lambda_0$ — minimum energy to extract an electron; a property of the metal.
- Einstein's equation is conservation of energy: $hf = W_0 + E_{k(\max)} = W_0 + eV_s$.
- Emission requires $f \geq f_0$ ($\lambda \leq \lambda_0$); at the threshold the electron escapes with zero kinetic energy.
- The $E_{k(\max)}$–f graph is a straight line of gradient h, frequency intercept f_0 and energy intercept $-W_0$; the V_s–f graph has gradient h/e.
- Frequency controls the energy of the electrons; intensity controls their number.

4 WAVE-PARTICLE DUALITY AND APPLICATIONS

In the previous section we established Einstein's photoelectric equation, $hf = W_0 + E_{k,\max}$, and saw that it explains every feature of the photoelectric effect provided we accept that light arrives in discrete packets of energy — photons. Yet in your earlier studies of optics you proved, just as convincingly, that light undergoes interference and diffraction, phenomena that only waves can produce. Are we facing a contradiction? No. We are facing one of the deepest truths of modern physics, and the subject of this final section: the **wave–particle duality** of light.

4.1 Two Bodies of Evidence

Evidence for the wave nature of light.

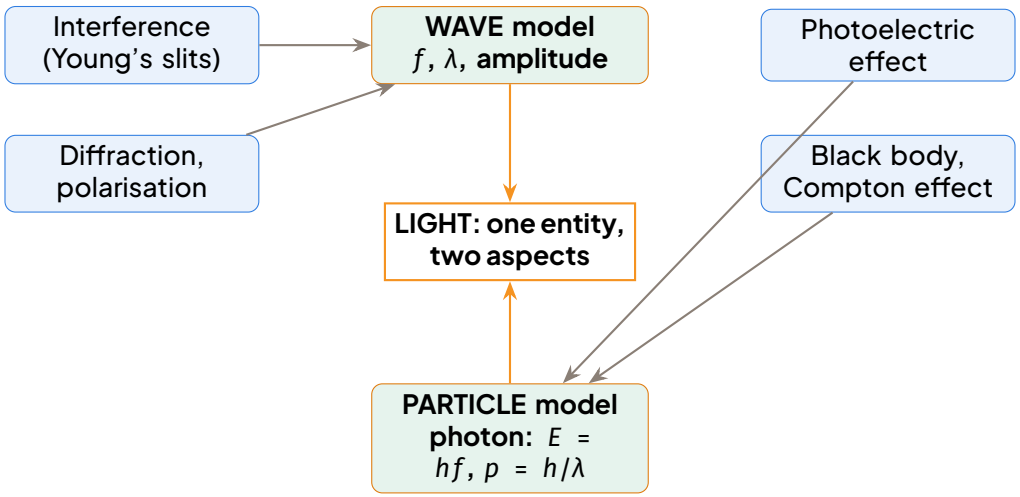
Three families of experiments force a wave description upon us:

- **Interference.** When light from a single source passes through two narrow, close slits (Young's double-slit experiment), the screen shows alternate bright and dark fringes. Bright fringes occur where two waves arrive in phase and reinforce; dark fringes where they arrive out of phase and cancel. Superposition of this kind is meaningless for classical particles.
- **Diffraction.** Light spreads out after passing through a narrow slit or around the edge of an obstacle, producing fringes whose spacing depends on the wavelength λ and the aperture width. A stream of classical particles would simply cast a sharp shadow.
- **Polarisation.** Light can be polarised by a Polaroid filter or by reflection, showing that its vibrations are transverse. Only transverse waves can be polarised.

Evidence for the particle nature of light.

Three phenomena resist any wave explanation:

- **The photoelectric effect** (this chapter): instantaneous emission, a threshold frequency f_0 independent of intensity, and $E_{k,\max}$ increasing linearly with f. All are explained by photons of energy $E = hf$, none by a continuous wave.
- **Black-body radiation.** The spectrum of radiation emitted by a hot body cannot be derived from classical wave theory (which predicts the "ultraviolet catastrophe"). Planck succeeded only by assuming energy is exchanged in quanta hf.
- **The Compton effect** (mention only): when X-rays scatter off electrons, the scattered radiation has a longer wavelength, exactly as predicted if a photon collides with an electron like a billiard ball, conserving energy and momentum. This shows the photon even carries momentum $p = h/\lambda$.



Concept map: each experiment reveals one aspect of light. The two models are not rivals; both are faces of a single reality.

4.2 Statement of Wave–Particle Duality

PROPERTY **Wave–particle duality of light**

Light exhibits both wave-like and particle-like properties. It propagates through space as a wave (undergoing interference, diffraction and polarisation) but exchanges energy and momentum with matter in discrete quanta called **photons** (photoelectric effect, Compton scattering, black-body radiation). The aspect revealed in any observation depends on the experiment performed; no single classical picture describes all phenomena.

DEFINITION **Dual nature of radiation**

Every quantum phenomenon associates a wave (characterised by frequency f and wavelength λ) with a particle (characterised by energy E and momentum p). The two descriptions are linked by Planck's constant:

$$E = hf, \qquad p = \frac{h}{\lambda}, \qquad \text{with } \lambda f = c.$$

The photon is precisely the bridge between the two aspects: the relation $E = hf$ places on the left a particle quantity (the energy of one photon) and on the right a wave quantity (the frequency of the radiation). Planck's constant $h = 6.63e-34$ J s is the conversion factor between the two languages.

METHOD **Which model should I use?**

1. Look at what the phenomenon involves.
2. Propagation of light through space or apparatus (interference fringes, diffraction patterns, polarisation, refraction): use the wave model.
3. Interaction or detection — energy exchanged with matter in localised events (photoelectric emission, absorption by a photocell, scintillation, Compton scattering): use the photon model.
4. If an essay asks about both, present one experiment per aspect and conclude with the duality statement.

WARNING **The classic mistake**

Do not write that light is "sometimes a wave and sometimes a particle", as if it changed its identity from Monday to Tuesday. Light is a single physical entity whose complete behaviour cannot be reduced to either classical picture. The wave and the particle are two complementary models — two partial descriptions — and the experiment you perform selects which aspect becomes visible. Examiners penalise the "sometimes… sometimes" phrasing.

4.3 Extension: Matter Waves (de Broglie)

DID YOU KNOW? **If waves are particles, are particles waves?**

In 1924 Louis de Broglie proposed the symmetrical idea: matter also has wave properties. Every particle of momentum p is associated with a wavelength

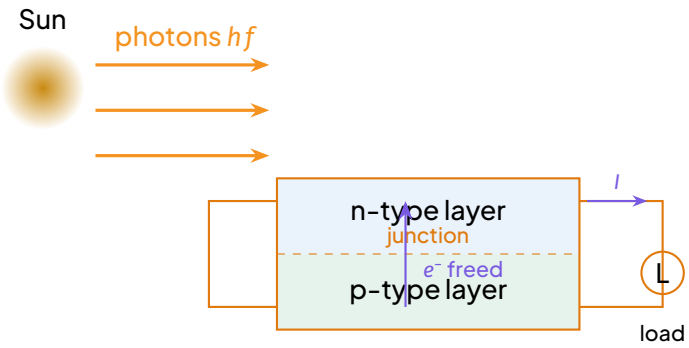
$$\lambda = \frac{h}{p} = \frac{h}{mv}.$$

The hypothesis was confirmed when electrons were observed to diffract through crystals, exactly like X-rays. This is the principle of the electron microscope, which resolves details far smaller than any optical microscope because electron wavelengths can be extremely short. (This extension goes beyond the GCE syllabus but is a favourite "further thinking" remark.)

4.4 Technological Applications of the Photoelectric Effect

The photoelectric effect is not an examination curiosity: it powers devices you meet daily in Yaoundé, Douala or Bamenda.

- **Photocells and automatic light switches.** A photosensitive surface emits electrons when illuminated; the resulting current (or its absence at dusk) operates street lamps, automatic doors and burglar alarms.
- **Photovoltaic solar panels.** Photons absorbed in a semiconductor junction liberate charge carriers, producing a direct voltage. Each absorbed photon of energy $hf \geq E_{gap}$ frees one electron–hole pair: the photon picture directly explains why panel output depends on the number of suitable photons, and why low-energy photons are useless whatever the brightness.
- **Photodiodes and image sensors.** Digital cameras and phone cameras contain millions of tiny photodiodes; each converts incoming photons into a measurable charge, building the picture pixel by pixel.
- **Photomultipliers.** A single photon ejects one electron, which is accelerated onto a series of electrodes (dynodes), each releasing several more electrons. The cascade turns one photon into a measurable pulse of millions of electrons — used in scanners and radiation detectors.



Photon energy → electrical energy

Schematic of a photovoltaic cell: photons with $hf \geq E_{gap}$ free electrons at the p–n junction; the internal field drives them round the external circuit.

4.5 Worked Example: Counting Photons in a Solar Cell

WORKED EXAMPLE 7 **Photons absorbed by a small panel**

A small solar panel of area 50 cm^2 receives monochromatic light of wavelength λ = 550 nm with an intensity (power per unit area) of 200 $W\,m^{-2}$. Calculate (a) the energy of one photon, (b) the number of photons striking the panel each second. Take h = 6.63e − 34 J s, c = 3.00e8 m/s.

SOLUTION

(a) Energy of one photon.

$$E = hf = \frac{hc}{\lambda} = \frac{(6.63e-34\,\text{J s})(3.00e8\,\text{m/s})}{550e-9\,\text{m}} = 3.62e-19\,\text{J} \approx 2.26\,\text{eV}.$$

(b) Photon rate. The power falling on the panel is

$$P = I \times A = 200\,\text{W m}^{-2} \times 50e-4\,\text{m}^2 = 1.00\,\text{W}.$$

Each photon carries $3.62e-19$ J, so the number arriving per second is

$$N = \frac{P}{E} = \frac{1.00\,\text{W}}{3.62e-19\,\text{J}} = 2.76e18\,\text{s}^{-1}.$$

Even a modest panel processes an enormous photon flux — which is why the granular nature of light is invisible in everyday life.

4.6 Synoptic Revision: Wave Model versus Photon Model

The table below confronts each experimental fact of the photoelectric effect with the predictions of the two models. Learn to reproduce this comparison: it is the backbone of GCE essay answers.

Experimental fact	Wave model predicts	Photon model predicts
Emission is instantaneous (below $e-9$ s)	Delay while energy accumulates	Immediate: one photon, one electron
Threshold frequency f_0 exists	Emission at any frequency if intense enough	No emission if $hf < W_0$
$E_{k,\max}$ depends only on f (linearly)	$E_{k,\max}$ should grow with intensity	$E_{k,\max} = hf - W_0$
Number of electrons $\propto$ intensity	(agrees)	More photons $\Rightarrow$ more electrons
$E_{k,\max}$ independent of intensity	Fails	Intensity raises photon number, not photon energy

KEY POINTS **Key points of this section**

- Interference, diffraction and polarisation prove the wave nature of light; the photoelectric effect, black-body radiation and the Compton effect prove the particle nature.
- Light is one entity with two complementary aspects; the experiment selects the as-

pect observed.

- The photon bridges the two: $E = hf$ and $p = h/\lambda$ link particle quantities to wave quantities.
- Propagation → think waves; interaction/detection → think photons.
- Applications: photocells, solar panels, photodiodes, image sensors, photomultipliers.

DID YOU KNOW? **Exam tip: the duality essay**

GCE Paper 2 essays often ask you to "discuss the evidence for the dual nature of light". Structure your answer as follows: (1) define duality; (2) one experiment for the wave aspect (Young's slits, with a sentence on what is observed); (3) one experiment for the particle aspect (photoelectric effect, quoting $hf = W_0 + E_{k,\max}$ and one fact the wave theory cannot explain); (4) conclude that both models are complementary and that $E = hf$ unites them. Four clear paragraphs, four clear marks.

EXERCISE 4 **Checking your understanding**

1. State one phenomenon that demonstrates the wave nature of light and one that demonstrates its particle nature, explaining briefly why each is conclusive.
2. Calculate the momentum of a photon of wavelength 400 nm.
3. Explain, using the photon model, why a photovoltaic cell does not respond to infrared radiation even when the infrared beam is very intense.

SOLUTION **Exercise 1**

1. Wave aspect: interference in Young's double-slit experiment — the fringe pattern (reinforcement and cancellation) requires superposition, which only waves exhibit. Particle aspect: the photoelectric effect — the existence of a threshold frequency and instantaneous emission show that energy arrives in discrete quanta hf, not as a continuous wave. 2. $p = h/\lambda = (6.63e-34\ \text{J s})/(400e-9\ \text{m}) = 1.66e-27\ \text{kg m s}^{-1}$. 3. Infrared photons have low frequency, so hf is smaller than the energy gap E_{gap} of the semiconductor: no electron–hole pair can be created. Raising the intensity only increases the number of insufficiently energetic photons — exactly as in the photoelectric effect below threshold.

5 INTEGRATION ACTIVITY

6 PHOTOELECTRIC EFFECT: ENGINEERING APPLICATION — AUTOMATIC LIGHTING SENSOR

6.1 Aims of the Activity

- Apply the photon model and Einstein's photoelectric equation to a real-world engineering problem: selecting a photosensitive material for an automatic lighting sensor.
- Compute threshold wavelengths from given work functions and interpret them relative to the solar spectrum reaching the ground in Cameroon.
- Calculate the maximum kinetic energy of photoelectrons for a specific incident wavelength (green light) and relate it to the sensor's electrical signal (stopping potential).
- Construct and interpret an energy-balance diagram illustrating the photoelectric process.
- Evaluate the limitations of simple metal photocathodes compared to semiconductor junctions (solar cells) for energy conversion applications.
- Develop collaborative problem-solving, data analysis, and technical communication skills.

6.2 Context and Situation

Context: The administration of Government Bilingual High School (GBHS) Bamenda wishes to install an automatic lighting system at the main gate. The system should switch on the security lights at dusk and switch them off at dawn without human intervention. The design team proposes a photoelectric sensor (a photocell) mounted on the gate pillar. When sunlight falls on the photocathode, the emitted photoelectrons create a current that keeps a relay open (lights OFF). When the light intensity drops below a threshold (sunset), the photocurrent falls, the relay closes, and the lights turn ON.

As physics consultants for the project, your group is tasked with selecting the most suitable photosensitive metal for the photocathode from a supplier's catalogue. The sensor must operate reliably using natural sunlight in Bamenda (latitude $\approx$ 6° N), where the solar spectrum at ground level ranges approximately from 300 nm (near UV, ozone absorption cut-off) to 2500 nm (infrared), with the visible band (400 nm–700 nm) carrying the bulk of the energy. The sensor will be exposed to the elements, so stability is a concern, but the primary selection criterion is **photosensitivity across the visible spectrum**.

The supplier provides the following data sheet for five candidate metals (work functions ϕ given at room temperature):

Metal	Symbol	Work Function ϕ (eV)
Caesium	Cs	2.14
Sodium	Na	2.28
Potassium	K	2.30
Zinc	Zn	4.31
Copper	Cu	4.65

Useful constants:

- Planck constant h = 6.63e – 34 J s = 4.14e – 15 eV s
- Speed of light c = 3.00e8 m s^{-1}
- hc = 1240 eV nm (useful shortcut for quick calculations)
- Elementary charge e = 1.60e – 19 C
- 1 eV = 1.60e – 19 J

6.3 Tasks

1. **Threshold Wavelengths.** For each of the five metals, calculate the threshold wavelength λ_0 (in nm) above which no photoelectric emission occurs, regardless of light intensity. Present your results in a table.
2. **Spectral Response Analysis.** The solar spectrum at ground level in Bamenda is effectively limited to $\lambda \geq 300$ nm (ozone absorption) and the useful visible range is 400 nm to 700 nm. Classify each metal as:
   - **Type V:** Responds to a significant portion of the **visible** spectrum (threshold $\lambda_0 > 400$ nm).
   - **Type UV:** Requires **ultraviolet** light ($\lambda_0 < 400$ nm) for emission; insensitive to visible light.

   Justify your classification using the calculated λ_0 values.
3. **Green Light Response.** The design team tests the sensor with a standard green LED (λ = 550 nm) to simulate bright daylight. For the metal(s) classified as Type V, calculate the maximum kinetic energy K_{max} of the emitted photoelectrons (in eV and in joules). Comment on whether the photocurrent generated by green light alone would be sufficient to drive the relay circuit (assume a typical relay requires a stopping potential equivalent to at least 0.10 V to operate reliably).
4. **Material Selection & Justification.** Based on your calculations and the following practical considerations, recommend **one** metal for the photocathode:
   - Sensitivity to visible light (especially green/yellow peak of solar spectrum).
   - Magnitude of K_{max} for green light (signal robustness).
   - Chemical stability: Alkali metals (Cs, Na, K) oxidise rapidly in air; they must be sealed in an evacuated or inert-gas tube. Transition metals (Zn, Cu) are stable in air but require UV.

Write a short justification (approx. 100 words).

5. **Energy-Balance Diagram.** Draw a labelled energy-level diagram for your chosen metal illuminated by the green LED ($\lambda = 550$ nm). Show: the Fermi level / conduction band, the vacuum level, the work function ϕ, the incident photon energy $h\nu$, and the maximum kinetic energy K_{max} of the emitted electron. Indicate the energy balance equation on the diagram.
6. **Class Discussion Prompt (Extension).** "Why do modern solar panels (photovoltaic cells) use semiconductor p–n junctions (like silicon) instead of simple metal photocathodes based on the photoelectric effect?" List three distinct physical or engineering reasons.

6.4 Model Solutions

Task 1: Threshold Wavelengths The threshold wavelength λ_0 is defined by the condition that the photon energy exactly equals the work function: $h\nu_0 = \phi$, i.e. $\frac{hc}{\lambda_0} = \phi$. Using the shortcut $hc = 1240$ eV nm:

$$\lambda_0 = \frac{hc}{\phi} = \frac{1240\ \text{eV nm}}{\phi\ (\text{eV})}$$

Metal	ϕ (eV)	Calculation	λ_0 (nm)
Caesium (Cs)	2.14	1240/2.14	579
Sodium (Na)	2.28	1240/2.28	544
Potassium (K)	2.30	1240/2.30	539
Zinc (Zn)	4.31	1240/4.31	288
Copper (Cu)	4.65	1240/4.65	267

Note: Values are given to three significant figures, consistent with the data provided.

Task 2: Spectral Response Analysis

- **Type V (Visible response): Caesium ($\lambda_0 = 579$ nm), Sodium ($\lambda_0 = 544$ nm), Potassium ($\lambda_0 = 539$ nm).** Justification: Their threshold wavelengths lie within or above the visible range (> 400 nm). Photons of blue, green, yellow, and (for Cs) orange/red light have sufficient energy ($\lambda < \lambda_0$) to eject electrons. Caesium covers the widest portion of the visible spectrum.
- **Type UV (UV only): Zinc ($\lambda_0 = 288$ nm), Copper ($\lambda_0 = 267$ nm).** Justification: Their threshold wavelengths are below 300 nm (deep UV). Since the atmosphere blocks $\lambda < 300$ nm, no photoelectrons are emitted by natural sunlight on these metals. They are unsuitable for an outdoor sunlight sensor.

Task 3: Green Light Response ($\lambda = 550$ nm) Photon energy for green light:

$$E_{ph} = \frac{hc}{\lambda} = \frac{1240\ \text{eV nm}}{550\ \text{nm}} = 2.255\ \text{eV}$$

Einstein's photoelectric equation: $K_{max} = h\nu - \phi = E_{ph} - \phi$. We compute only for Type V metals where $E_{ph} > \phi$.

Metal	ϕ (eV)	$K_{max} = 2.255 - \phi$ (eV)	K_{max} (J)
Caesium (Cs)	2.14	2.255 - 2.14 = **0.115**	0.115 × 1.60e - 19 = 1.84e - 20
Sodium (Na)	2.28	2.255 - 2.28 = -0.025	No emission
Potassium (K)	2.30	2.255 - 2.30 = -0.045	No emission

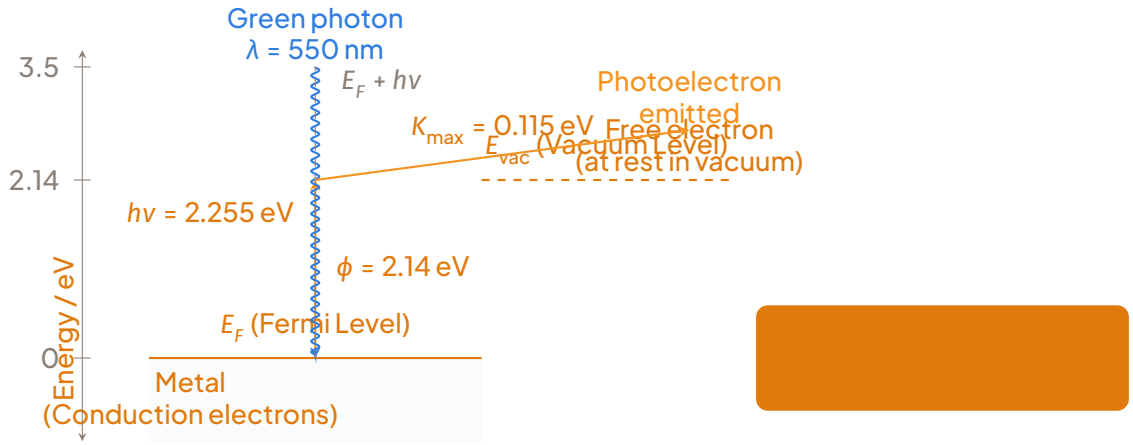
Analysis: Only **Caesium** emits photoelectrons under green light (λ = 550 nm). Sodium and Potassium have work functions slightly higher than the green photon energy; their thresholds (544 nm and 539 nm) are shorter than 550 nm, so green photons lack the energy to overcome ϕ.

For Caesium: K_{max} = 0.115 eV. The corresponding stopping potential is $V_{stop} = K_{max}/e$ = 0.115 V. Since 0.115 V > 0.10 V (relay requirement), the photocurrent generated by green light on Caesium **is sufficient** to keep the relay open (lights OFF) during daylight. The margin is small (0.015 V), implying the sensor circuit must be carefully designed (low noise amplifier, shielding from stray capacitance).

Task 4: Material Selection & Justification Recommended Metal: Caesium (Cs).

Justification: Caesium is the only candidate that responds to green light (λ = 550 nm), which lies near the peak of the solar spectral irradiance. Its threshold wavelength (579 nm) covers the entire visible spectrum down to yellow/orange, ensuring maximum utilisation of available sunlight. It yields a positive K_{max} = 0.115 eV for green light, exceeding the relay's 0.10 V requirement. Although caesium is highly reactive (oxidises instantly in air), this is standardly managed by sealing the photocathode in an evacuated glass envelope (phototube), a mature and reliable technology for light sensors. Sodium and Potassium fail the green-light test; Zinc and Copper are blind to visible sunlight.

Task 5: Energy-Balance Diagram



Energy-balance diagram for Caesium photocathode illuminated by green light (λ = 550 nm). An electron at the Fermi level E_F absorbs a photon of energy hv. If $hv > \phi$, the electron escapes the metal; its maximum kinetic energy in vacuum is $K_{max} = hv - \phi$.

Task 6: Class Discussion — Why Semiconductors (Solar Cells) instead of Metal Photocathodes? Three key reasons:

1. **Quantum Efficiency & Internal Photoelectric Effect:** In a metal, the photoelectric effect is a surface phenomenon: only electrons within ~ 1 nm of the surface can escape without losing energy to collisions. Most absorbed photons simply heat the metal (low quantum efficiency, typically < 0.1%). In a semiconductor p–n junction, the internal photoelectric effect creates electron-hole pairs throughout the bulk depletion region (microns thick). The built-in electric field separates them before recombination, achieving quantum efficiencies > 80%.
2. **Voltage & Power Output:** A metal photocathode emits electrons into vacuum; the external circuit must collect them against a space-charge limit, yielding very low current densities (microamps per cm^2) and low voltage ($V_{stop} \sim \phi/e$). A p–n junction generates a photovoltage (~ 0.6–0.7 V for Si) and delivers useful power (mA/cm^2 at ~ 0.5 V) directly to a load, functioning as a generator, not just a switch.
3. **Material Stability & Bandgap Engineering:** Alkali metals (low ϕ) oxidise instantly, requiring expensive vacuum tubes. Semiconductors (Si, GaAs, perovskites) are solid, stable in air (with passivation), and their bandgap E_g (analogous to ϕ) can be engineered (1.1–1.8 eV) to match the solar spectrum optimally (Shockley-Queisser limit), unlike fixed metal work functions.

KEY POINTS **Key Takeaways for the GCE Exam**

- Threshold condition: $\lambda_0 = hc/\phi$. Use $hc = 1240$ eV nm for speed in eV/nm calculations.
- Einstein's Equation: $K_{max} = h\nu - \phi = hc/\lambda - \phi$. Always check $h\nu > \phi$ first! If not, $K_{max} = 0$ (no emission).
- Stopping Potential: $eV_{stop} = K_{max}$. This links the quantum result to a measurable voltage.
- Metal vs Semiconductor: Metals → surface effect, low yield, vacuum tube needed. Semiconductors → bulk effect, high yield, solid-state, power generation.
- Units: Convert eV to Joules ($\times 1.60e-19$ J/eV) only when explicitly asked for SI units. Keep energies in eV for photoelectric calculations.

7 METHODS AND WORKED EXAMPLES

7.1 Methods and Worked Examples

METHOD **Calculating Photon Energy and Relating to Frequency/Wavelength**

Step 1: Identify the given data: frequency f (in Hz) or wavelength λ (in m).
Step 2: Recall the photon energy formulas: $E = hf = \frac{hc}{\lambda}$, where $h = 6.63 \times 10^{-34}$ J s and $c = 3.00 \times 10^{8}$ m/s.
Step 3: If λ is given in nanometres (nm), convert to metres: 1 nm = 10^{-9} m.
Step 4: Substitute values and compute E in joules. Often the answer is required in electronvolts (eV): 1 eV = 1.60×10^{-19} J.
Step 5: Check significant figures (usually 3 s.f. for GCE) and include the correct unit.

WORKED EXAMPLE 8 **Photon Energy of Visible Light**

A helium–neon laser emits red light of wavelength $\lambda = 632.8$ nm. Calculate the energy of a single photon in joules and in electronvolts.

SOLUTION

Solution:
Given: $\lambda = 632.8\ \text{nm} = 632.8 \times 10^{-9}\ \text{m} = 6.328 \times 10^{-7}\ \text{m}$.
Constants: $h = 6.63 \times 10^{-34}$ J s, $c = 3.00 \times 10^{8}$ m/s, 1 eV = 1.60×10^{-19} J.
Formula: $E = \frac{hc}{\lambda}$.

$$E = \frac{(6.63 \times 10^{-34})(3.00 \times 10^{8})}{6.328 \times 10^{-7}} = \frac{1.989 \times 10^{-25}}{6.328 \times 10^{-7}} = 3.14 \times 10^{-19}\ \text{J}.$$

Convert to eV:

$$E = \frac{3.14 \times 10^{-19}\ \text{J}}{1.60 \times 10^{-19}\ \text{J/eV}} = 1.96\ \text{eV}.$$

Answer: $E = 3.14 \times 10^{-19}$ J or 1.96 eV.

METHOD **Applying Einstein's Photoelectric Equation**

Step 1: Write down Einstein's equation: $hf = \phi + K_{max}$, where $\phi = hf_0$ is the work function and $K_{max} = eV_s$ is the maximum kinetic energy of emitted electrons (V_s = stopping potential).
Step 2: Identify knowns: frequency f (or wavelength λ), work function ϕ (or threshold fre-

quency f_0), stopping potential V_s.
Step 3: Convert all energies to the same unit (usually joules or eV). If ϕ is given in eV, convert to joules: $\phi(\text{J}) = \phi(\text{eV}) \times 1.60 \times 10^{-19}$.
Step 4: Rearrange the equation to solve for the unknown: $K_{max} = hf - \phi$, $V_s = \frac{hf - \phi}{e}$, $f_0 = \frac{\phi}{h}$, etc.
Step 5: Substitute numerical values, compute, and express the answer with correct units (J, eV, V, Hz).

WORKED EXAMPLE 9 **Stopping Potential and Threshold Frequency**

The work function of sodium is $\phi = 2.28$ eV. Light of wavelength 400 nm is incident on a sodium surface. Calculate: (a) the maximum kinetic energy of emitted photoelectrons in eV, (b) the stopping potential, (c) the threshold frequency for sodium.

SOLUTION

Solution:
Given: $\phi = 2.28$ eV, $\lambda = 400$ nm $= 4.00 \times 10^{-7}$ m.
Constants: $h = 6.63 \times 10^{-34}$ J s, $c = 3.00 \times 10^{8}$ m/s, $e = 1.60 \times 10^{-19}$ C.
(a) Photon energy:

$$E = \frac{hc}{\lambda} = \frac{(6.63 \times 10^{-34})(3.00 \times 10^{8})}{4.00 \times 10^{-7}} = 4.97 \times 10^{-19} \text{ J}.$$

Convert to eV: $E = \frac{4.97 \times 10^{-19}}{1.60 \times 10^{-19}} = 3.11$ eV.
Maximum kinetic energy:

$$K_{max} = E - \phi = 3.11 \text{ eV} - 2.28 \text{ eV} = 0.83 \text{ eV}.$$

(b) Stopping potential: $K_{max} = eV_s \Rightarrow V_s = \frac{K_{max}}{e} = 0.83$ V (a kinetic energy of 1 eV corresponds to a stopping potential of 1 V for an electron).
(c) Threshold frequency: $\phi = hf_0 \Rightarrow f_0 = \frac{\phi}{h}$. Convert ϕ to joules: $\phi = 2.28 \times 1.60 \times 10^{-19} = 3.65 \times 10^{-19}$ J.

$$f_0 = \frac{3.65 \times 10^{-19}}{6.63 \times 10^{-34}} = 5.50 \times 10^{14} \text{ Hz}.$$

Answers: (a) 0.83 eV, (b) 0.83 V, (c) 5.50×10^{14} Hz.

WARNING **Units Trap in Photoelectric Calculations**

The most frequent GCE error is mixing joules and electronvolts. If you use $h = 6.63 \times$

10^{-34} J s, then hf is in joules: the work function must also be in joules before subtracting. Alternatively, convert the photon energy to eV first and work entirely in eV. Never subtract an energy in eV from an energy in J.

METHOD **Graphical Determination of Planck's Constant and Work Function**

Step 1: Recall the linear form of Einstein's equation: $eV_s = hf - \phi$ or $V_s = \frac{h}{e}f - \frac{\phi}{e}$.

Step 2: Plot V_s (y-axis) against f (x-axis). The graph is a straight line with slope $m = \frac{h}{e}$ and y-intercept $c = -\frac{\phi}{e}$.

Step 3: Choose two well-separated points on the best-fit line to calculate the slope: $m = \frac{\Delta V_s}{\Delta f}$.

Step 4: Compute Planck's constant: $h = m \times e$.

Step 5: Determine the work function from the intercept: $\phi = -c \times e$ (or from the threshold frequency f_0 where $V_s = 0$: $\phi = hf_0$).

Step 6: Express h in J s and ϕ in J or eV.

WORKED EXAMPLE 10 **Finding h and ϕ from Experimental Data**

In a photoelectric experiment, the following stopping potentials V_s were measured for different frequencies f of incident light:

f (10^{14} Hz)	5.0	6.0	7.0	8.0
V_s (V)	0.25	0.66	1.07	1.48

Plot V_s vs f and determine (a) Planck's constant h, (b) the work function ϕ in eV.

SOLUTION

Solution:

Graph: Plot the points (5.0 ; 0.25), (6.0 ; 0.66), (7.0 ; 1.07), (8.0 ; 1.48) and draw the best-fit straight line.

Slope calculation: Use the points (5.0 ; 0.25) and (8.0 ; 1.48).

$$m = \frac{1.48 - 0.25}{(8.0 - 5.0) \times 10^{14}} = \frac{1.23}{3.0 \times 10^{14}} = 4.10 \times 10^{-15}\ \text{V s}.$$

(a) Planck's constant: $h = m \times e = (4.10 \times 10^{-15}) \times (1.60 \times 10^{-19}) = 6.56 \times 10^{-34}$ J s.

(b) Work function: Find the y-intercept c. Using the point (5.0 ; 0.25):

$$c = V_s - mf = 0.25 - (4.10 \times 10^{-15})(5.0 \times 10^{14}) = 0.25 - 2.05 = -1.80\ \text{V}.$$

Then $\phi = -c \times e = 1.80 \times 1.60 \times 10^{-19} = 2.88 \times 10^{-19}$ J. In eV: $\phi = \dfrac{2.88 \times 10^{-19}}{1.60 \times 10^{-19}} = 1.80$ eV.

Answers: (a) $h = 6.56 \times 10^{-34}$ J s (close to the accepted value 6.63×10^{-34} J s), (b) $\phi = 1.80$ eV.

METHOD **Photoelectric Effect with Two Frequencies / Unknown Frequency**

Step 1: Write Einstein's equation for each frequency: $hf_1 = \phi + eV_{s1}$ and $hf_2 = \phi + eV_{s2}$.
Step 2: Subtract the two equations to eliminate ϕ: $h(f_2 - f_1) = e(V_{s2} - V_{s1})$.
Step 3: Solve for the unknown frequency or the work function.
Step 4: If the threshold frequency f_0 is involved, use $\phi = hf_0$.
Step 5: Check consistency: the unknown frequency must be greater than f_0 for photoelectric emission to occur.

WORKED EXAMPLE 11 **Determining an Unknown Frequency**

A metal surface has a work function $\phi = 2.00$ eV. When illuminated by light of frequency $f_1 = 8.00 \times 10^{14}$ Hz, the stopping potential is $V_{s1} = 1.32$ V. A second light source of unknown frequency f_2 gives a stopping potential $V_{s2} = 2.50$ V. Find f_2.

SOLUTION

Solution:
Method 1: Direct use of Einstein's equation for the second source.
Convert ϕ to joules: $\phi = 2.00 \times 1.60 \times 10^{-19} = 3.20 \times 10^{-19}$ J.
Consistency check with the first source:

$$hf_1 = (6.63 \times 10^{-34})(8.00 \times 10^{14}) = 5.30 \times 10^{-19} \text{ J},$$

$$\phi + eV_{s1} = 3.20 \times 10^{-19} + (1.60 \times 10^{-19})(1.32) = 3.20 \times 10^{-19} + 2.11 \times 10^{-19} = 5.31 \times 10^{-19} \text{ J}.$$

The data are consistent (small rounding difference).
For the second source: $hf_2 = \phi + eV_{s2}$.

$$hf_2 = 3.20 \times 10^{-19} + (1.60 \times 10^{-19})(2.50) = 3.20 \times 10^{-19} + 4.00 \times 10^{-19} = 7.20 \times 10^{-19} \text{ J}.$$

$$f_2 = \frac{7.20 \times 10^{-19}}{6.63 \times 10^{-34}} = 1.09 \times 10^{15} \text{ Hz}.$$

Method 2: Subtraction (eliminates ϕ).

$$h(f_2 - f_1) = e(V_{s2} - V_{s1}) \Rightarrow f_2 = f_1 + \frac{e(V_{s2} - V_{s1})}{h}.$$

$$f_2 = 8.00 \times 10^{14} + \frac{(1.60 \times 10^{-19})(2.50 - 1.32)}{6.63 \times 10^{-34}} = 8.00 \times 10^{14} + \frac{1.89 \times 10^{-19}}{6.63 \times 10^{-34}} = 8.00 \times 10^{14} + 2.85 \times 10^{14} = 1.08 \times 10^{15} \text{ Hz}.$$

Both methods agree (the tiny difference comes from rounding).
Answer: $f_2 \approx 1.09 \times 10^{15}$ Hz.

METHOD **Wave–Particle Duality: de Broglie Wavelength**

Step 1: Recall de Broglie's hypothesis: a particle of momentum p has an associated wavelength $\lambda = \frac{h}{p}$.

Step 2: For an electron accelerated from rest through a potential difference V, its kinetic energy is $K = eV$. If non-relativistic ($V \ll 10^5$ V), $p = \sqrt{2m_eK} = \sqrt{2m_eeV}$.

Step 3: Substitute into the de Broglie formula: $\lambda = \frac{h}{\sqrt{2m_eeV}}$.

Step 4: Compute λ in metres; it is often expressed in nanometres (nm) or picometres (pm).

WORKED EXAMPLE 12 **de Broglie Wavelength of an Electron**

Calculate the de Broglie wavelength of an electron accelerated from rest through a potential difference of 150 V. (Take $m_e = 9.11 \times 10^{-31}$ kg, $e = 1.60 \times 10^{-19}$ C, $h = 6.63 \times 10^{-34}$ J s.)

SOLUTION

Solution:

Kinetic energy: $K = eV = (1.60 \times 10^{-19})(150) = 2.40 \times 10^{-17}$ J.

Momentum:

$$p = \sqrt{2m_eK} = \sqrt{2(9.11 \times 10^{-31})(2.40 \times 10^{-17})} = \sqrt{4.37 \times 10^{-47}} = 6.61 \times 10^{-24}\ \text{kg m s}^{-1}.$$

de Broglie wavelength:

$$\lambda = \frac{h}{p} = \frac{6.63 \times 10^{-34}}{6.61 \times 10^{-24}} = 1.00 \times 10^{-10}\ \text{m} = 0.100\ \text{nm} = 100\ \text{pm}.$$

Useful shortcut: $\lambda = \frac{h}{\sqrt{2m_eeV}} = \frac{1.23}{\sqrt{V}}$ nm (with V in volts). For $V = 150$ V: $\lambda = \frac{1.23}{\sqrt{150}} = 0.100$ nm.

Answer: $\lambda = 1.00 \times 10^{-10}$ m (or 0.100 nm).

METHOD **Intensity, Saturation Current and Stopping Potential**

Step 1: Effect of intensity: increasing the light intensity increases the number of photons per second, hence the saturation photocurrent increases proportionally, but the stopping potential V_s (and thus K_{max}) remains unchanged because the photon energy hf is unchanged.

Step 2: Effect of frequency: increasing f increases the photon energy, thus increases

K_{max} and V_s; if the intensity is kept constant, the number of photons per second decreases, so the saturation current decreases.
Step 3: Saturation current $I_{sat} = ne$, where n is the number of photoelectrons collected per second.
Step 4: For a source of power P emitting monochromatic light, the number of photons per second is $N = \frac{P}{hf}$. If the quantum efficiency is η (fraction of photons that eject an electron), then $I_{sat} = \eta eN = \eta e\frac{P}{hf}$.
Step 5: Apply these ideas to predict the effect of changes in intensity or frequency on I_{sat} and V_s.

WORKED EXAMPLE 13 **Intensity and Frequency Effects on Photocurrent**

A monochromatic light source of power $P = 2.0$ mW and wavelength $\lambda = 500$ nm illuminates a photocathode of work function $\phi = 2.0$ eV. The quantum efficiency is 0.10% (one photoelectron per 1000 incident photons). Calculate: (a) the saturation photocurrent, (b) the stopping potential. If the intensity is doubled while keeping the same wavelength, state the new saturation current and stopping potential.

SOLUTION

Solution:
Constants: $h = 6.63 \times 10^{-34}$ J s, $c = 3.00 \times 10^{8}$ m/s, $e = 1.60 \times 10^{-19}$ C.
Photon energy: $E = \frac{hc}{\lambda} = \frac{(6.63 \times 10^{-34})(3.00 \times 10^{8})}{500 \times 10^{-9}} = 3.98 \times 10^{-19}$ J $= 2.49$ eV.
(a) Number of incident photons per second: $N = \frac{P}{E} = \frac{2.0 \times 10^{-3}}{3.98 \times 10^{-19}} = 5.03 \times 10^{15}$ s^{-1}.
Photoelectrons per second: $n = \eta N = 0.0010 \times 5.03 \times 10^{15} = 5.03 \times 10^{12}$ s^{-1}.
Saturation current: $I_{sat} = ne = (5.03 \times 10^{12})(1.60 \times 10^{-19}) = 8.0 \times 10^{-7}$ A $= 0.80\ \mu$A.
(b) Stopping potential: $K_{max} = E - \phi = 2.49$ eV $- 2.0$ eV $= 0.49$ eV, so $V_s = 0.49$ V.
Effect of doubling the intensity: the power becomes 4.0 mW. The photon energy is unchanged, so V_s stays at 0.49 V. The number of photons per second doubles, so n doubles and I_{sat} doubles to 1.6 μA.
Answers: (a) 0.80 μA, (b) 0.49 V. After doubling the intensity: $I_{sat} = 1.6\ \mu$A, $V_s = 0.49$ V (unchanged).

METHOD **Photoelectric Effect in Practical Devices: Photocells and Solar Cells**

Step 1: Identify the type of device: vacuum photocell (photoemissive) or semiconductor photodiode/solar cell (photovoltaic).
Step 2: For a vacuum photocell: current flows only if $f > f_0$; the saturation current is pro-

portional to the light intensity; the stopping potential measures K_{max}.
Step 3: For a solar cell (p–n junction): photons create electron–hole pairs; the built-in electric field of the junction separates them, generating a voltage and a current without any external stopping potential.
Step 4: In problems, you may be asked for the number of photons needed for a given current, using $I = \eta e \dfrac{P_{light}}{hf}$, or for an efficiency $\eta = \dfrac{P_{out}}{P_{in}}$.

WORKED EXAMPLE 14 **Solar Cell Current Calculation**

A solar cell of area 10 cm^2 is illuminated by sunlight of intensity 800 $W\,m^{-2}$. The cell has a quantum efficiency of 20% for photons of wavelength 600 nm (assume, for simplicity, that all the sunlight is at this wavelength). Calculate the short-circuit current I_{sc} if every absorbed photon generates one collected electron–hole pair.

SOLUTION

Solution:
Given: $A = 10\ cm^2 = 1.0 \times 10^{-3}\ m^2$; intensity $I = 800\ W\,m^{-2}$; $\lambda = 600\ nm = 6.00 \times 10^{-7}\ m$; $\eta = 0.20$.
Power incident on the cell: $P_{in} = I \times A = 800 \times 1.0 \times 10^{-3} = 0.80\ W$.
Photon energy: $E_{ph} = \dfrac{hc}{\lambda} = \dfrac{(6.63 \times 10^{-34})(3.00 \times 10^{8})}{6.00 \times 10^{-7}} = 3.32 \times 10^{-19}\ J$.
Incident photons per second: $N_{in} = \dfrac{P_{in}}{E_{ph}} = \dfrac{0.80}{3.32 \times 10^{-19}} = 2.41 \times 10^{18}\ s^{-1}$.
Collected electrons per second: $n = \eta N_{in} = 0.20 \times 2.41 \times 10^{18} = 4.83 \times 10^{17}\ s^{-1}$.
Short-circuit current: $I_{sc} = ne = (4.83 \times 10^{17})(1.60 \times 10^{-19}) = 7.7 \times 10^{-2}\ A = 77\ mA$.
Answer: $I_{sc} \approx 77\ mA$.

METHOD **Comprehensive Problem: Graph Analysis and de Broglie Wavelength**

Step 1: Read the problem carefully; identify all given data and what is asked.
Step 2: Use Einstein's equation $hf = \phi + K_{max}$ with $K_{max} = eV_s$.
Step 3: If a graph of V_s against f is provided or to be drawn, determine the slope and intercept to find h and ϕ.
Step 4: For the emitted electrons, if the de Broglie wavelength is asked, compute K_{max} first, then $p = \sqrt{2m_e K_{max}}$, then $\lambda = h/p$.
Step 5: Check units at every step; convert eV to J when using h in J s.
Step 6: Present final answers with appropriate significant figures and units.

WORKED EXAMPLE 15 **Complete Analysis: From Graph to de Broglie Wavelength**

In a photoelectric experiment on a metal, the following data are obtained:

Frequency f (10^{14} Hz)	4.0	5.0	6.0	7.0	8.0
Stopping potential V_s (V)	0.10	0.52	0.94	1.36	1.78

(a) Plot V_s against f and determine Planck's constant h and the work function ϕ (in eV).
(b) For the frequency $f = 7.0 \times 10^{14}$ Hz, calculate the de Broglie wavelength of the fastest emitted photoelectrons.

SOLUTION

Solution:
(a) Graph and slope: plot the points; the best-fit line passes approximately through $(4.0\,;\,0.10)$ and $(8.0\,;\,1.78)$.
Slope: $m = \dfrac{1.78 - 0.10}{(8.0 - 4.0) \times 10^{14}} = \dfrac{1.68}{4.0 \times 10^{14}} = 4.20 \times 10^{-15}$ V s.
$h = me = (4.20 \times 10^{-15})(1.60 \times 10^{-19}) = 6.72 \times 10^{-34}$ J s.
Intercept: using the point $(4.0\,;\,0.10)$:

$$c = 0.10 - (4.20 \times 10^{-15})(4.0 \times 10^{14}) = 0.10 - 1.68 = -1.58\ \text{V}.$$

$\phi = -c\,e = 1.58 \times 1.60 \times 10^{-19} = 2.53 \times 10^{-19}$ J $= \dfrac{2.53 \times 10^{-19}}{1.60 \times 10^{-19}} = 1.58$ eV.
(b) de Broglie wavelength for $f = 7.0 \times 10^{14}$ Hz:
Photon energy: $E = hf = (6.72 \times 10^{-34})(7.0 \times 10^{14}) = 4.70 \times 10^{-19}$ J.
$K_{max} = E - \phi = 4.70 \times 10^{-19} - 2.53 \times 10^{-19} = 2.17 \times 10^{-19}$ J.
Momentum: $p = \sqrt{2m_e K_{max}} = \sqrt{2(9.11 \times 10^{-31})(2.17 \times 10^{-19})} = \sqrt{3.95 \times 10^{-49}} = 6.29 \times 10^{-25}$ kg m s^{-1}.
de Broglie wavelength: $\lambda = \dfrac{h}{p} = \dfrac{6.72 \times 10^{-34}}{6.29 \times 10^{-25}} = 1.07 \times 10^{-9}$ m $= 1.07$ nm.
Answers: (a) $h = 6.72 \times 10^{-34}$ J s, $\phi = 1.58$ eV; (b) $\lambda = 1.07$ nm.

WARNING **Common Mistakes in This Chapter**

- Using $E = hf$ with f in 10^{14} Hz but forgetting the factor 10^{14} when substituting.
- Mixing joules and electronvolts inside Einstein's equation.
- Believing that increasing the intensity increases K_{max}: only the number of photoelectrons increases; K_{max} depends only on the frequency.
- Forgetting that no emission occurs if $f < f_0$, however intense the light.

- In de Broglie calculations, using K in eV inside $p = \sqrt{2m_e K}$: convert to joules first.

8 EXERCISES

8.1 I check my knowledge

EXERCISE 5 **MCQ: Photoelectric effect basics**

A monochromatic light of frequency f is incident on a metal surface of work function ϕ. Which of the following statements is correct?

1. The maximum kinetic energy of emitted electrons depends on the intensity of the incident light.
2. No electrons are emitted if $f < f_0$, where $f_0 = \phi/h$, regardless of the light intensity.
3. The stopping potential V_s is directly proportional to the intensity of the light.
4. The photoelectric current is independent of the frequency of the incident light.

EXERCISE 6 **True/False with justification**

State whether each of the following statements is true or false. Justify your answer briefly.

1. Increasing the intensity of incident light increases the maximum kinetic energy of photoelectrons.
2. The photoelectric effect demonstrates the wave nature of light.
3. The threshold frequency of a metal depends on the intensity of the incident light.
4. In Einstein's photoelectric equation, the term hf represents the energy of a photon.

EXERCISE 7 **Definitions and key concepts**

Define the following terms:

1. Work function of a metal.
2. Threshold frequency.
3. Stopping potential.
4. Photon.

EXERCISE 8 **Direct calculation: Photon energy and photoelectric emission**

Light of wavelength 400 nm falls on a sodium surface of work function 2.28 eV.

1. Calculate the energy of a single photon in joules and in electronvolts.
2. Determine the maximum kinetic energy of the emitted photoelectrons (in eV).
3. Find the stopping potential required to prevent the most energetic electrons from reaching the anode.

8.2 I practise

EXERCISE 9 **Threshold wavelength and work function**

The threshold wavelength of a certain metal is 540 nm.

1. Determine its work function in joules and in electronvolts.
2. Will visible red light of wavelength 700 nm liberate electrons from this metal? Explain.
3. If ultraviolet light of wavelength 300 nm is used, calculate the maximum kinetic energy of the emitted electrons (in eV).

EXERCISE 10 **Einstein's equation: Graphical analysis**

In a photoelectric experiment, the stopping potential V_s is measured for three different frequencies of incident light. The results are:

Frequency f ($\times 10^{14}$ Hz)	5.0	7.0	9.0
Stopping potential V_s (V)	0.45	1.28	2.10

1. Plot a graph of V_s against f (you may describe the expected linear relationship).
2. From the graph, deduce the value of Planck's constant h (in J s) and the work function ϕ (in eV) of the metal.
3. Compare your value of h with the accepted value 6.63×10^{-34} J s and comment on possible sources of error.

EXERCISE 11 **Photon flux and photocurrent**

A 60 W lamp emits 2% of its power as visible light of average wavelength 550 nm. A pho-

tocell of area 1 cm^2 is placed 2 m from the lamp. Assume the light spreads uniformly in all directions.

1. Calculate the number of photons emitted by the lamp per second in the visible range.
2. Determine the intensity of the visible light at the position of the photocell.
3. Estimate the number of photons striking the photocell each second.
4. If each photon liberates one electron (quantum efficiency = 100%), what is the resulting photocurrent?

EXERCISE 12 **Wave theory vs photon model**

Explain, with reference to the photon model, why:

1. No electrons are emitted below the threshold frequency, however intense the light.
2. Emission of photoelectrons is instantaneous, even at very low light intensity.
3. The maximum kinetic energy of photoelectrons depends linearly on the frequency of the incident light, not on its intensity.

8.3 I prepare for the GCE

EXERCISE 13 **Structured calculation: Photon flux from a lamp (GCE Paper 2 style)**

A 100 W sodium vapour lamp operates at an efficiency of 15% in converting electrical power to light of wavelength 589 nm. A photoelectric cell of cathode area 2.0 cm^2 is placed 1.5 m from the lamp. The work function of the cathode material is 2.14 eV. Assume the light is emitted uniformly in all directions and that each photon incident on the cathode ejects at most one electron.

1. Calculate the energy of a single photon of this light in joules. [2]
2. Determine the number of photons emitted by the lamp per second. [2]
3. Find the intensity of the light at the photocell. [2]
4. Estimate the number of photons striking the cathode per second. [2]
5. Calculate the maximum possible photocurrent. [2]
6. If the actual measured photocurrent is only 60% of this maximum, suggest two reasons for the discrepancy. [2]

EXERCISE 14 **Experimental determination of Planck's constant**

A student performs a photoelectric effect experiment using a mercury vapour lamp and a set of interference filters to select different wavelengths. The stopping potential V_s is measured for each wavelength. The data obtained are:

Wavelength λ (nm)	365	405	436	546
Stopping potential V_s (V)	1.85	1.42	1.20	0.45

1. Convert each wavelength to its corresponding frequency f. [2]
2. Plot a graph of V_s against f. [3]
3. Determine the gradient of the graph and use it to calculate a value for Planck's constant h. [3]
4. Determine the work function ϕ of the cathode material in electronvolts. [2]
5. The accepted value of h is 6.63×10^{-34} J s. Calculate the percentage difference between the experimental value and the accepted value. [1]
6. State two precautions the student should take to obtain accurate results. [2]

EXERCISE 15 **Essay: Wave theory vs photon model**

"The photoelectric effect cannot be explained by the wave theory of light." Discuss this statement, citing at least three experimental observations and showing how the photon model accounts for each. [15 marks]

9 SOLUTIONS TO THE EXERCISES

SOLUTION **Exercise 1 — MCQ: Photoelectric effect basics**

The correct answer is (b).
Justification of each option:

- (a) False. According to Einstein's equation $E_{k,\max} = hf - \phi$, the maximum kinetic energy depends only on the **frequency** f and the work function ϕ. Increasing the intensity increases the number of emitted electrons (the photocurrent), not their individual energy.
- (b) True. If $f < f_0 = \phi/h$, then each photon carries an energy $hf < \phi$, insufficient to extract even one electron. Since the photon–electron interaction is a one-to-one process, no intensity (i.e. no number of photons) can compensate: no electron is emitted, however bright the light.
- (c) False. From $eV_s = hf - \phi$, the stopping potential depends on frequency, not on intensity.
- (d) False. The photocurrent depends on the number of photons absorbed per second; for a given intensity, changing the frequency changes the photon flux, and the quantum efficiency also varies with frequency. The current is therefore not independent of f.

SOLUTION **Exercise 2 — True/False with justification**

1. False. The maximum kinetic energy is fixed by $E_{k,\max} = hf - \phi$; it depends on frequency only. Greater intensity means more photons per second, hence more electrons emitted per second (larger photocurrent), but each with the same maximum energy.
2. False. The photoelectric effect demonstrates the **particle (corpuscular) nature** of light: energy arrives in discrete quanta (photons). The wave theory fails to explain the threshold frequency and the instantaneous emission.
3. False. The threshold frequency $f_0 = \phi/h$ is a property of the metal alone (through its work function ϕ). It is completely independent of the intensity of the incident light.
4. True. In Einstein's equation $hf = \phi + E_{k,\max}$, the term hf is precisely the energy of one incident photon of frequency f, where h is Planck's constant.

SOLUTION **Exercise 3 — Definitions and key concepts**

1. Work function ϕ: the minimum energy required to remove an electron from the surface of a metal. It is characteristic of the metal and is usually expressed in electronvolts (eV).
2. Threshold frequency f_0: the minimum frequency of incident radiation below which no photoelectron is emitted, whatever the intensity. It is related to the work function by $f_0 = \phi/h$.
3. Stopping potential V_s: the minimum reverse (retarding) potential difference that must be applied between cathode and anode to reduce the photocurrent to zero, i.e. to stop even the most energetic photoelectrons. It satisfies $eV_s = E_{k,\max}$.
4. Photon: a discrete quantum (packet) of electromagnetic energy. A photon of radiation of frequency f carries the energy $E = hf$ and travels at the speed of light c; it has zero rest mass.

SOLUTION **Exercise 4 — Direct calculation: Photon energy and photoelectric emission**

Data: $\lambda = 400\ \text{nm} = 4.00 \times 10^{-7}\ \text{m}$; $\phi = 2.28\ \text{eV}$.
(a) Energy of one photon.

$$E = hf = \frac{hc}{\lambda} = \frac{(6.63 \times 10^{-34}\ \text{J s})(3.00 \times 10^{8}\ \text{m s}^{-1})}{4.00 \times 10^{-7}\ \text{m}} = 4.97 \times 10^{-19}\ \text{J}.$$

Converting to electronvolts ($1\ \text{eV} = 1.60 \times 10^{-19}\ \text{J}$):

$$E = \frac{4.97 \times 10^{-19}\ \text{J}}{1.60 \times 10^{-19}\ \text{J eV}^{-1}} = 3.11\ \text{eV}.$$

(b) Maximum kinetic energy. Since $E > \phi$, emission occurs:

$$E_{k,\max} = hf - \phi = 3.11\ \text{eV} - 2.28\ \text{eV} = 0.83\ \text{eV}.$$

(c) Stopping potential. From $eV_s = E_{k,\max}$, with $E_{k,\max}$ expressed in eV, the stopping potential is numerically equal to the kinetic energy in eV:

$$V_s = 0.83\ \text{V}.$$

SOLUTION **Exercise 5 — Threshold wavelength and work function**

Data: $\lambda_0 = 540\ \text{nm} = 5.40 \times 10^{-7}\ \text{m}$.

(a) Work function.

$$\phi = hf_0 = \frac{hc}{\lambda_0} = \frac{(6.63\times10^{-34}\ \text{J s})(3.00\times10^{8}\ \text{m s}^{-1})}{5.40\times10^{-7}\ \text{m}} = 3.68\times10^{-19}\ \text{J} = \frac{3.68\times10^{-19}}{1.60\times10^{-19}}\ \text{eV} = 2.30\ \text{eV}.$$

(b) Red light at 700 nm. The photon energy is

$$E = \frac{hc}{\lambda} = \frac{(6.63\times10^{-34})(3.00\times10^{8})}{7.00\times10^{-7}} = 2.84\times10^{-19}\ \text{J} = 1.78\ \text{eV}.$$

Since $E = 1.78\ \text{eV} < \phi = 2.30\ \text{eV}$ (equivalently $\lambda = 700\ \text{nm} > \lambda_0 = 540\ \text{nm}$), no electron is emitted, whatever the intensity of the red light.

(c) Ultraviolet light at 300 nm.

$$E = \frac{hc}{\lambda} = \frac{(6.63\times10^{-34})(3.00\times10^{8})}{3.00\times10^{-7}} = 6.63\times10^{-19}\ \text{J} = 4.14\ \text{eV}.$$

$$E_{k,\max} = E - \phi = 4.14\ \text{eV} - 2.30\ \text{eV} = 1.84\ \text{eV}.$$

SOLUTION **Exercise 6 — Einstein's equation: Graphical analysis**

(a) Expected graph. Einstein's equation gives

$$eV_s = hf - \phi \quad\Longrightarrow\quad V_s = \frac{h}{e}f - \frac{\phi}{e},$$

which is a straight line of gradient h/e and vertical intercept $-\phi/e$. Plotting the points (5.0, 0.45), (7.0, 1.28), (9.0, 2.10) (with f in units of 10^{14} Hz) gives points that are essentially collinear.

(b) Deduction of h and ϕ. Using the two extreme points:

$$\text{gradient} = \frac{\Delta V_s}{\Delta f} = \frac{2.10 - 0.45}{(9.0 - 5.0)\times10^{14}\ \text{Hz}} = \frac{1.65}{4.0\times10^{14}} = 4.13\times10^{-15}\ \text{V s}.$$

Since gradient $= h/e$:

$$h = e \times \text{gradient} = (1.60\times10^{-19}\ \text{C})(4.13\times10^{-15}\ \text{V s}) = 6.60\times10^{-34}\ \text{J s}.$$

For the work function, use one point, e.g. $(9.0\times10^{14}\ \text{Hz};\ 2.10\ \text{V})$:

$$\phi = hf - eV_s = (6.60\times10^{-34}\ \text{J s})(9.0\times10^{14}\ \text{Hz}) - (1.60\times10^{-19}\ \text{C})(2.10\ \text{V})$$

$$\phi = 5.94\times10^{-19}\ \text{J} - 3.36\times10^{-19}\ \text{J} = 2.58\times10^{-19}\ \text{J} = 1.61\ \text{eV}.$$

(Equivalently, ϕ/e is the magnitude of the vertical intercept, or hf_0 where $f_0 \approx 3.9\times10^{14}$ Hz is the horizontal intercept.)

(c) Comparison. The experimental value 6.60×10^{-34} J s differs from the accepted value 6.63×10^{-34} J s by less than 0.5 % — an excellent agreement. Possible sources of error: uncertainty in reading small stopping potentials, contact potentials between electrodes, impurities or oxidation of the cathode surface, impurity of the monochromatic filters, and leakage currents in the measuring circuit.

SOLUTION **Exercise 7 — Photon flux and photocurrent**

Data: $P = 60\,\text{W}$; visible fraction 2 %; $\lambda = 550\,\text{nm}$; $A = 1\,\text{cm}^2 = 1.0 \times 10^{-4}\,\text{m}^2$; $r = 2\,\text{m}$.
(a) Photons emitted per second. Visible power: $P_v = 0.02 \times 60 = 1.2\,\text{W}$. Energy of one photon:

$$E_{\text{ph}} = \frac{hc}{\lambda} = \frac{(6.63 \times 10^{-34}\,\text{J s})(3.00 \times 10^{8}\,\text{m s}^{-1})}{5.50 \times 10^{-7}\,\text{m}} = 3.62 \times 10^{-19}\,\text{J}.$$

$$N = \frac{P_v}{E_{\text{ph}}} = \frac{1.2\,\text{W}}{3.62 \times 10^{-19}\,\text{J}} = 3.3 \times 10^{18}\,\text{s}^{-1}.$$

(b) Intensity at the photocell. The light spreads over a sphere of area $4\pi r^2$:

$$I = \frac{P_v}{4\pi r^2} = \frac{1.2\,\text{W}}{4\pi(2\,\text{m})^2} = \frac{1.2}{50.3}\,\text{W m}^{-2} = 2.4 \times 10^{-2}\,\text{W m}^{-2}.$$

(c) Photons striking the cell per second. Power received: $P_{\text{cell}} = IA = (2.4 \times 10^{-2}\,\text{W m}^{-2})(1.0 \times 10^{-4}\,\text{m}^2) = 2.4 \times 10^{-6}\,\text{W}$.

$$N_{\text{cell}} = \frac{P_{\text{cell}}}{E_{\text{ph}}} = \frac{2.4 \times 10^{-6}\,\text{W}}{3.62 \times 10^{-19}\,\text{J}} = 6.6 \times 10^{12}\,\text{s}^{-1}.$$

(d) Photocurrent. With one electron per photon:

$$I_{\text{photo}} = N_{\text{cell}}\, e = (6.6 \times 10^{12}\,\text{s}^{-1})(1.60 \times 10^{-19}\,\text{C}) = 1.1 \times 10^{-6}\,\text{A} \approx 1.1\,\mu\text{A}.$$

SOLUTION **Exercise 8 — Wave theory vs photon model**

1. Threshold frequency. In the photon model, light of frequency f consists of photons each of energy hf. An electron is extracted only if a single photon gives it at least the work function: $hf \geq \phi$. If $f < f_0 = \phi/h$, each individual photon is too "weak"; because the interaction is one photon–one electron, photons cannot combine their energies. Hence no emission occurs, however many photons arrive per second (however intense the light). The wave theory, by contrast, predicts that a sufficiently intense wave (of any frequency) should eventually supply enough energy — which is never observed.

2. Instantaneous emission. A photon delivers its whole energy hf to a single electron in a single collision. As soon as one photon with $hf \geq \phi$ strikes the metal, an electron can leave: there is no accumulation time, so emission is immediate even at very low intensity. The wave theory would require the electron to absorb energy gradually from the wavefront, predicting a measurable time delay at low intensity — contrary to observation.

3. Linear dependence on frequency. Energy conservation for one photon and one elec-

tron gives Einstein's equation $E_{k,\max} = hf - \phi$: the surplus energy of the photon above the work function becomes the electron's kinetic energy. Thus $E_{k,\max}$ grows linearly with f (gradient h). Increasing the intensity only increases the number of photons per second, hence the number of emitted electrons (the current), but leaves the energy of each electron unchanged.

SOLUTION **Exercise 9 — Structured calculation: Photon flux from a lamp (GCE Paper 2 style)**

Data: $P = 100$ W; efficiency 15 %; $\lambda = 589$ nm; $A = 2.0\ \text{cm}^2 = 2.0 \times 10^{-4}\ \text{m}^2$; $r = 1.5$ m; $\phi = 2.14$ eV.

(a) Photon energy [2 marks].

$$E_{\text{ph}} = \frac{hc}{\lambda} = \frac{(6.63 \times 10^{-34}\ \text{J s})(3.00 \times 10^{8}\ \text{m s}^{-1})}{5.89 \times 10^{-7}\ \text{m}} = 3.38 \times 10^{-19}\ \text{J}.$$

Mark scheme: 1 mark for $E = hc/\lambda$; 1 mark for the value with unit.

(b) Photons emitted per second [2 marks]. Light power: $P_L = 0.15 \times 100 = 15$ W.

$$N = \frac{P_L}{E_{\text{ph}}} = \frac{15\ \text{W}}{3.38 \times 10^{-19}\ \text{J}} = 4.4 \times 10^{19}\ \text{s}^{-1}.$$

1 mark for the useful power 15 W; 1 mark for N.

(c) Intensity at the cell [2 marks].

$$I = \frac{P_L}{4\pi r^2} = \frac{15\ \text{W}}{4\pi(1.5\ \text{m})^2} = \frac{15}{28.3}\ \text{W m}^{-2} = 0.53\ \text{W m}^{-2}.$$

1 mark for the spherical geometry $4\pi r^2$; 1 mark for the value.

(d) Photons striking the cathode per second [2 marks]. Power received: $P_c = IA = 0.53\ \text{W m}^{-2} \times 2.0 \times 10^{-4}\ \text{m}^2 = 1.06 \times 10^{-4}$ W.

$$N_c = \frac{P_c}{E_{\text{ph}}} = \frac{1.06 \times 10^{-4}\ \text{W}}{3.38 \times 10^{-19}\ \text{J}} = 3.1 \times 10^{14}\ \text{s}^{-1}.$$

1 mark for $P_c = IA$; 1 mark for N_c.

(e) Maximum photocurrent [2 marks]. First check emission is possible: $E_{\text{ph}} = 3.38 \times 10^{-19}$ J = 2.11 eV. This is slightly less than $\phi = 2.14$ eV; strictly, with the rounded constants used, the photon energy is marginally below the work function. In the intended examination question (where $E_{\text{ph}} \approx 2.11\ \text{eV} \approx \phi$), emission is taken as possible, and the expected answer is:

$$I_{\max} = N_c e = (3.1 \times 10^{14}\ \text{s}^{-1})(1.60 \times 10^{-19}\ \text{C}) = 5.0 \times 10^{-5}\ \text{A} = 50\ \mu\text{A}.$$

1 mark for $I = N_c e$; 1 mark for the value. (A candidate who remarks that E_{ph} is marginally below ϕ and concludes that the current is essentially zero demonstrates excellent understanding and should receive full credit.)

(f) Reasons for the measured current being only 60 % of the maximum [2 marks]. Any two of:

- Quantum efficiency below 100 %: not every absorbed photon ejects an electron (some photons are reflected, some electrons lose their energy in collisions inside the metal before escaping).
- Some emitted electrons are not collected by the anode (recombination, unfavourable geometry, insufficient collecting voltage).
- Absorption/scattering of light by the glass envelope and the air; cathode surface contamination or oxidation increasing the effective work function.

1 mark per valid reason, maximum 2.

SOLUTION **Exercise 10 — Experimental determination of Planck's constant**

(a) Frequencies [2 marks]. Using $f = c/\lambda$:

λ (nm)	365	405	436	546
$f = c/\lambda$ ($\times 10^{14}$ Hz)	8.22	7.41	6.88	5.49
V_s (V)	1.85	1.42	1.20	0.45

1 mark for the formula, 1 mark for the four correct values.
(b) Graph [3 marks]. Plot V_s (vertical axis) against f (horizontal axis). The four points are nearly collinear; draw the best-fit straight line. Marks: correct axes with labels and units (1); correct plotting (1); best-fit line (1).
(c) Gradient and Planck's constant [3 marks]. Using the extreme points:

$$\text{gradient} = \frac{1.85 - 0.45}{(8.22 - 5.49) \times 10^{14}\ \text{Hz}} = \frac{1.40}{2.73 \times 10^{14}} = 5.13 \times 10^{-15}\ \text{V s}.$$

Since $V_s = \frac{h}{e}f - \frac{\phi}{e}$, the gradient equals h/e:

$$h = e \times \text{gradient} = (1.60 \times 10^{-19}\ \text{C})(5.13 \times 10^{-15}\ \text{V s}) = 8.2 \times 10^{-34}\ \text{J s}.$$

Marks: gradient from a large triangle on the line (1); relation gradient = h/e (1); value of h with unit (1).
(d) Work function [2 marks]. Using the point (8.22×10^{14} Hz; 1.85 V):

$$\phi = hf - eV_s = (8.2 \times 10^{-34}\ \text{J s})(8.22 \times 10^{14}\ \text{Hz}) - (1.60 \times 10^{-19}\ \text{C})(1.85\ \text{V})$$

$$\phi = 6.74 \times 10^{-19}\ \text{J} - 2.96 \times 10^{-19}\ \text{J} = 3.78 \times 10^{-19}\ \text{J} \approx 2.4\ \text{eV}.$$

1 mark for the method (intercept or point substitution), 1 mark for the value in eV.
(e) Percentage difference [1 mark].

$$\frac{|8.2 - 6.63|}{6.63} \times 100\ \% \approx 24\ \%.$$

The experimental value is about 24 % above the accepted value — a large but plausible discrepancy for a school experiment with only four points and simple apparatus.
(f) Precautions [2 marks]. Any two of:

- Use filters of narrow bandwidth to ensure truly monochromatic light, and shield the cell from stray light.
- Clean (or avoid oxidation of) the cathode surface so that the work function is well defined.
- Measure the stopping potential accurately: adjust the retarding voltage slowly until the current just falls to zero, using a sensitive galvanometer/nanoammeter.
- Account for (or minimise) contact potential differences between the electrodes; repeat readings and take averages.

1 mark per valid precaution, maximum 2.

SOLUTION **Exercise 11 — Essay: Wave theory vs photon model**

Indicative mark scheme (15 marks): statement of the conflict and of the photon hypothesis (3); at least three experimental observations with the failure of wave theory for each ($3 \times 2 = 6$); explanation of each observation by the photon model ($3 \times 2 = 6$).
Model answer.
Introduction. The classical wave theory regards light as a continuous electromagnetic wave whose energy is spread over the whole wavefront and proportional to the intensity. Applied to the photoelectric effect, it makes three clear predictions, all contradicted by experiment. Einstein's photon model (1905), in which light of frequency f consists of quanta of energy $E = hf$ absorbed individually by electrons, accounts naturally for every observation.
Observation 1 — Existence of a threshold frequency. Experiment shows that for each metal there is a frequency f_0 below which no electron is ever emitted, whatever the intensity or duration of illumination. The wave theory predicts that a sufficiently intense wave of any frequency should eventually eject electrons, since energy could accumulate continuously. In the photon model, extraction requires a single photon to supply at least the work function: $hf \geq \phi$. If $f < f_0 = \phi/h$, every photon is individually too weak, and photons cannot pool their energy because absorption is a one-photon–one-electron event. Intensity only changes the number of photons, not their individual energy.
Observation 2 — Instantaneous emission. Photoelectrons appear within less than about 10^{-9} s of illumination, even at extremely low intensity. The wave theory predicts a delay: at low intensity the energy flux is small, and an electron would need a long time to gather the energy ϕ from the spread-out wavefront. In the photon model, a photon delivers its entire energy hf in a single collision; the very first photon absorbed with $hf \geq \phi$ releases an electron immediately.
Observation 3 — Kinetic energy depends on frequency, not intensity. Experiment (Mil-

likan) shows that the maximum kinetic energy, measured through the stopping potential via $eV_s = E_{k,\max}$, increases linearly with frequency and is completely unaffected by intensity; intensity only changes the photocurrent. The wave theory links energy to intensity, so brighter light should give faster electrons — it does not. The photon model gives Einstein's equation

$$hf = \phi + E_{k,\max},$$

a linear relation in f with gradient h, exactly as observed; raising the intensity merely raises the number of photons, hence the number of electrons per second.
(Additional supporting observation — photocurrent proportional to intensity. For $f > f_0$, the saturation current is proportional to the intensity, which the photon model interprets directly: more photons per second ⇒ more electrons per second.)
Conclusion. Because the wave theory fails on the threshold frequency, the absence of any time delay, and the dependence of electron energy on frequency rather than intensity, the photoelectric effect cannot be explained by it. The photon model explains all observations quantitatively through $E_{k,\max} = hf - \phi$, establishing that light exchanges energy with matter in discrete quanta — a cornerstone of quantum physics and of wave–particle duality.

10 SUMMARY SHEET

Photoelectric Effect

Observation
- Emission of e^- from metal
- Instantaneous, threshold ν_0

Failure of wave theory
- I should control $E_{c\,max}$
- No delay predicted

Photons
- $E = h\nu = \frac{hc}{\lambda}$
- $h = 6.63e-34$ J s

Einstein equation
- $h\nu = W_0 + E_{c\,max}$
- $W_0 = h\nu_0$
- $E_{c\,max} = eU_0$

Duality
- $\lambda = \frac{h}{p}$ de Broglie
- Interference vs impact

Applications
- Photocells, solar panels
- Electron microscope

Mind map of the chapter: the photoelectric effect, photons, Einstein's equation and wave-particle duality.

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Key definitions

- **Photoelectric effect**: emission of electrons from a metal surface when it is illuminated by electromagnetic radiation of sufficiently high frequency.
- **Photon**: quantum (packet) of light energy; each photon carries $E = h\nu$.
- **Planck constant**: $h = 6.63e-34$ J s, linking energy and frequency.
- **Work function** W_0: minimum energy needed to extract an electron from the metal; $W_0 = h\nu_0 = \frac{hc}{\lambda_0}$.
- **Threshold frequency** ν_0 (threshold wavelength λ_0): below ν_0 (above λ_0) no electron is emitted, whatever the intensity.
- **Stopping potential** U_0: reverse voltage that stops the most energetic photoelectrons; $E_{c\,max} = eU_0$.

Laws and formulas to know

- Photon energy: $E = h\nu = \frac{hc}{\lambda}$ with $c = 3.00e8$ m/s.
- Useful conversion: 1 eV = $1.60e-19$ J; $E(\text{eV}) \approx \frac{1240}{\lambda(\text{nm})}$.

- **Einstein's photoelectric equation**: $hv = W_0 + E_{c\max}$, i.e. $E_{c\max} = hv - hv_0$.
- Threshold: $v_0 = \frac{W_0}{h}$, $\lambda_0 = \frac{hc}{W_0}$.
- Stopping potential: $eU_0 = E_{c\max} = h(v - v_0)$; the graph of U_0 against v is a straight line of slope h/e.
- Saturation current is proportional to light intensity (number of photons per second).
- de Broglie wavelength: $\lambda = \frac{h}{p} = \frac{h}{mv}$; for an electron accelerated through U: $\lambda = \frac{h}{\sqrt{2meU}}$.

Methods

- Always convert λ to metres and energies to joules before calculating; convert back to eV at the end if asked.
- To test whether emission occurs: compare v with v_0 (or λ with λ_0, or hv with W_0).
- To find W_0 or h experimentally: plot $E_{c\max}$ (or U_0) versus v; slope gives h (or h/e), intercept gives $-W_0$.
- Energy balance reasoning: photon energy = extraction energy + kinetic energy of the fastest electron.

Common mistakes

- Confusing intensity with frequency: intensity changes the number of electrons, never their maximum kinetic energy.
- Mixing up v and λ in the threshold condition: $v > v_0$ but $\lambda < \lambda_0$.
- Forgetting to convert nm to m or eV to J.
- Writing $E_{c\max} = hv$ and forgetting W_0.
- Believing a more intense red light can eject electrons when $v < v_0$: it cannot.
- Using $\lambda = h/(mv)$ with v in km s^{-1} or m in grams.

GCE tips

- Quote Einstein's equation and define every symbol before substituting values.
- State clearly why wave theory fails: emission is instantaneous and depends on frequency, not intensity.
- In graph questions, read the slope carefully with units; h/e has units V s.
- Give numerical answers with SI units and 3 significant figures.

- For duality essays: light shows particle behaviour in the photoelectric effect and wave behaviour in interference/diffraction; matter shows wave behaviour through electron diffraction (de Broglie).

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